



Fundamentals of Optics

Lecturer

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Q&A time: 2-4 pm on Monday

Room 239, South Tower, College of Engineering

Rules

- Final exam: 40%
- Mid-term exam: 30%
- Homework: 20%
- Attendance: 10%

Reference books

Optics, by Eugene Hecht
光学, 赵凯华、钟锡华

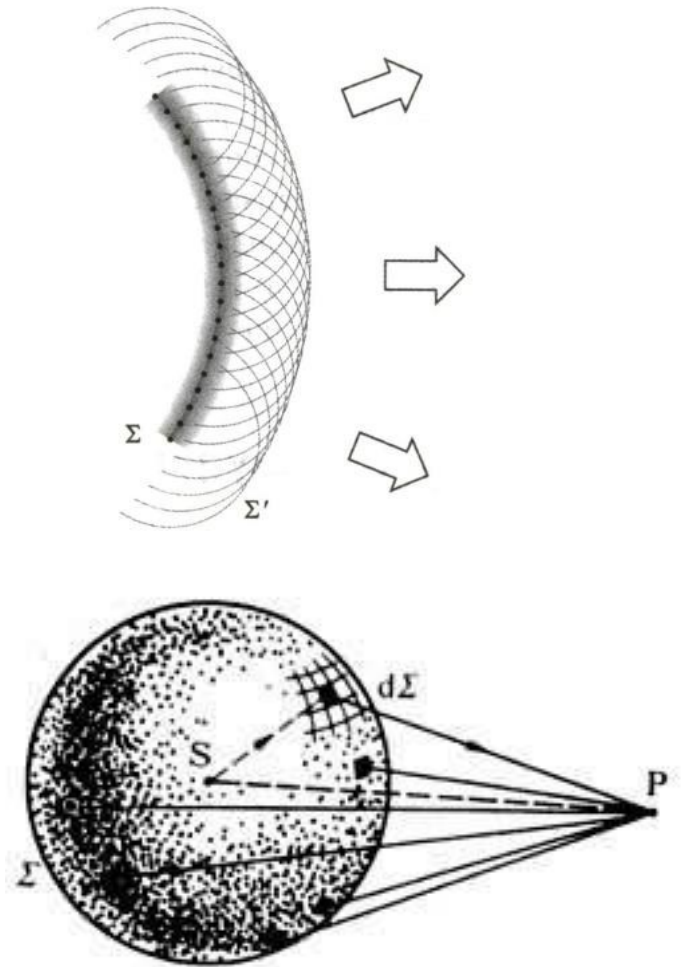
CHAPTER 9

DIFFRACTION

9.1 The Huygens-Fresnel Principle

Huygens's principle: every point on a propagating wavefront serves as the source of spherical secondary wavelets, such that the wavefront at some later time is the envelope of these wavelets.

Huygens-Fresnel principle: Every unobstructed point of a wavefront, at a given instant, serves as a source of spherical secondary wavelets (with the same frequency as that of the primary wave). The amplitude of the optical field at any point beyond is the superposition of all these wavelets (considering their amplitudes and relative phases)

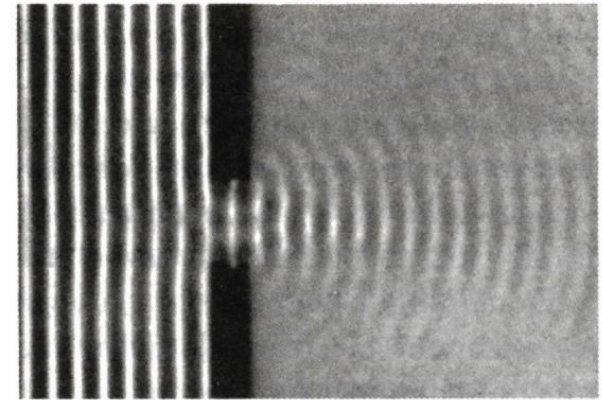
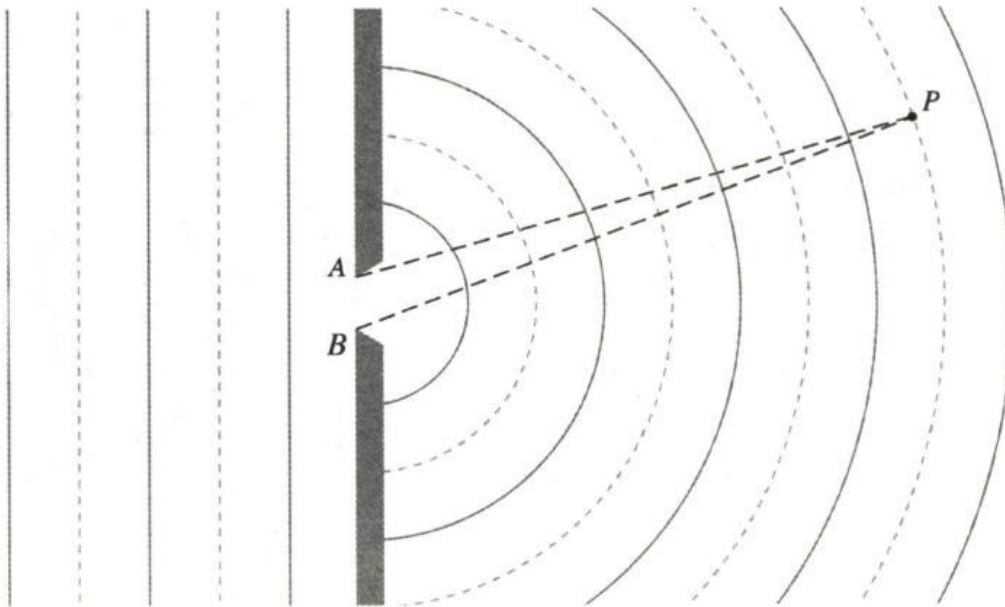


Example: ripple tank photograph:

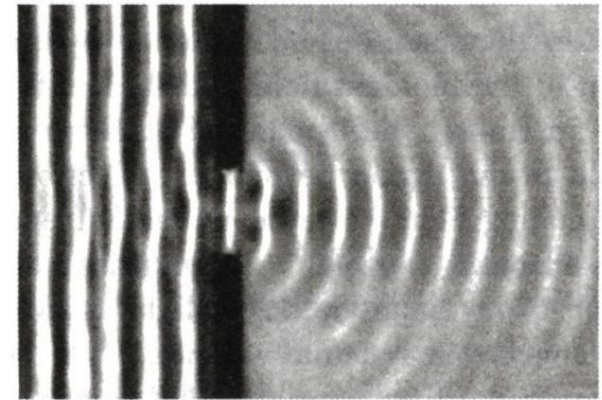
The maximum optical path length difference:

$$\Lambda_{max} = |AP - AB|$$

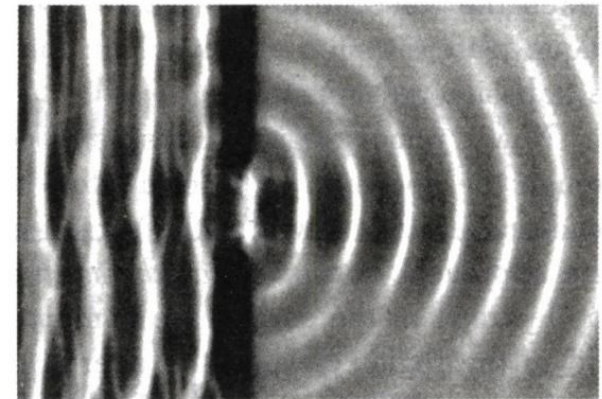
- I. $\lambda > AB$, namely $\lambda > \Lambda_{max}$: all interference is constructive whenever P happens to be.
- II. $\lambda < AB$: all the wavelets will interfere constructively in a small region where $\lambda > \Lambda_{max}$. Beyond this zone, some of wavelets can interfere destructively.



(a)

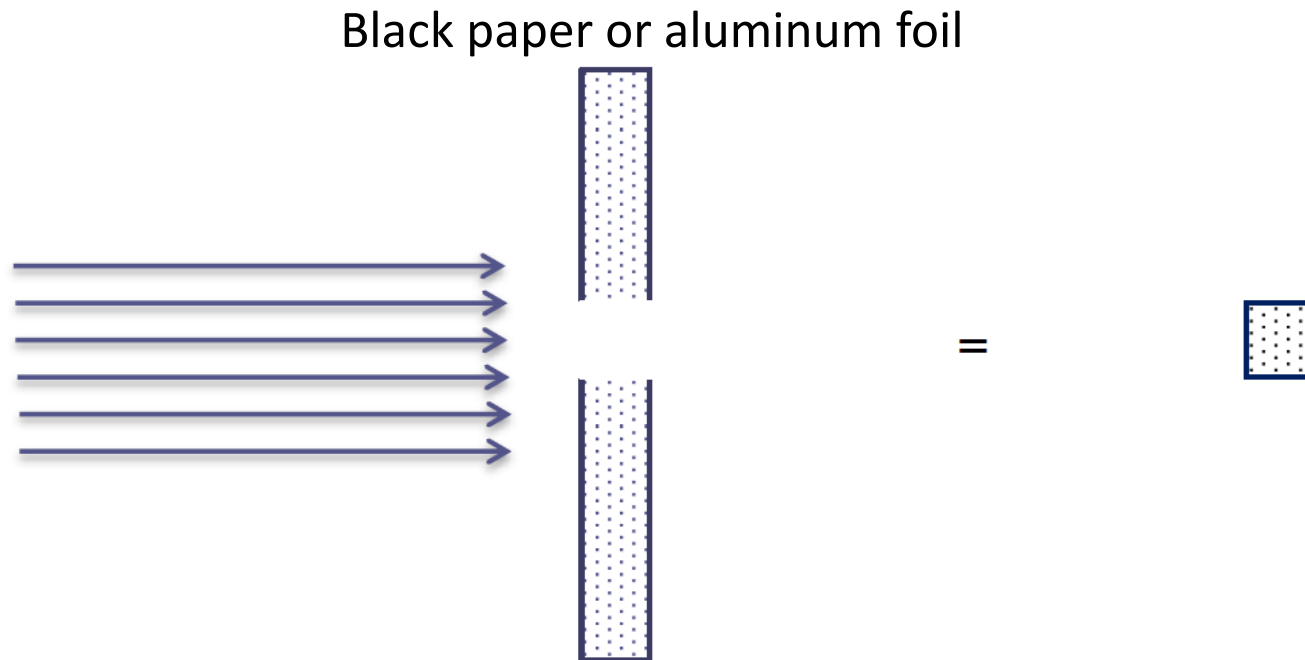


(b)



(c)

9.1.1 Opaque Obstructions



The diffraction field can be pictured as arising exclusively from a set of fictitious noninteracting oscillators distributed uniformly over the region of the aperture --- **Huygens-Fresnel Principle**.

9.1.2 Fraunhofer and Fresnel Diffraction

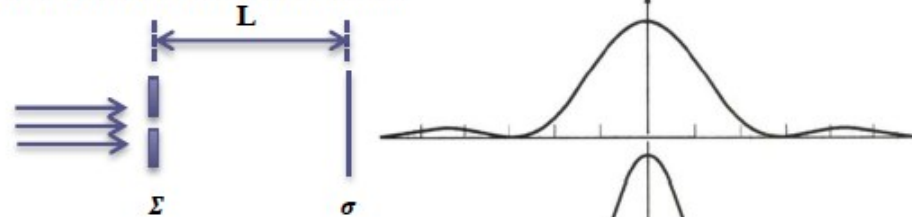
Fraunhofer (far-field) diffraction: at a long distance, the projected pattern have spread out considerably, bearing little or no resemblance to the actual aperture.

As long as both the incoming and outgoing waves approach being planar over the extent of the diffraction apertures, **Fraunhofer diffraction** obtains.

Fresnel (near-field) diffraction: the image of the aperture, though still easily recognizable, becomes increasingly more structured as the fringes become more prominent.

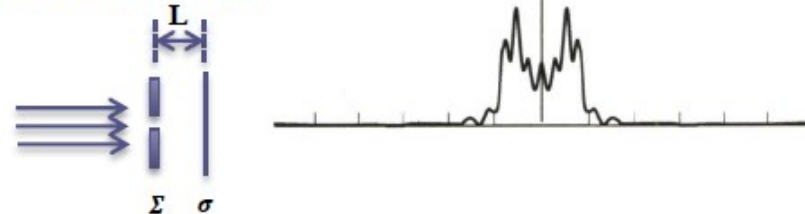
When the position of observation plane (σ) is fixed and a point source is moved toward (Σ), spherical waves would impinge on the aperture, and the **Fresnel pattern** would exist.

Fraunhofer (far-field) diffraction



$L \nearrow$

Fresnel (near-field) diffraction



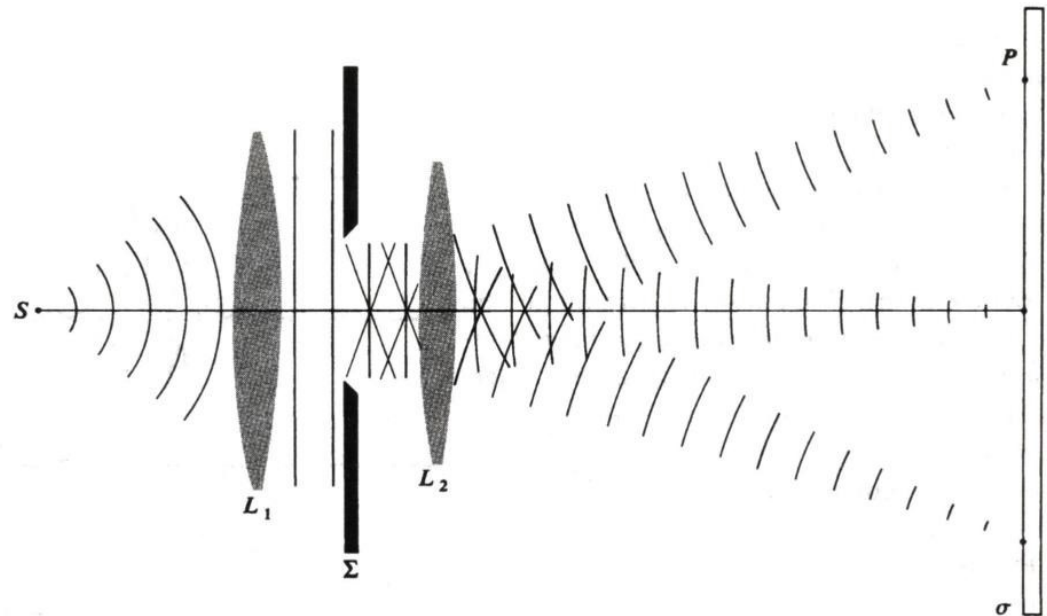
Fraunhofer diffraction: the wavefronts impinging on, and emerging from the aperture are planar. And the observation is at infinity. So the path difference could be described by a linear function of the two aperture variables.

Rule of thumb

$$R > a^2/\lambda$$

a : the width of aperture.

R : the smaller one of the two distance from S to Σ and Σ to P .



Achieve Fraunhofer diffraction using lenses

Fresnel diffraction: a spherical wavefront illuminates the hole, and the distance from S to each point on the screen is different and field strength is also different. Much the same thing is true for the diffracted waves going from the aperture to P .

9.1.3 Several Coherent Oscillators

Condition: a linear array of N coherent point oscillator with no intrinsic phase difference.

The separate wave amplitudes arriving at P is equal:

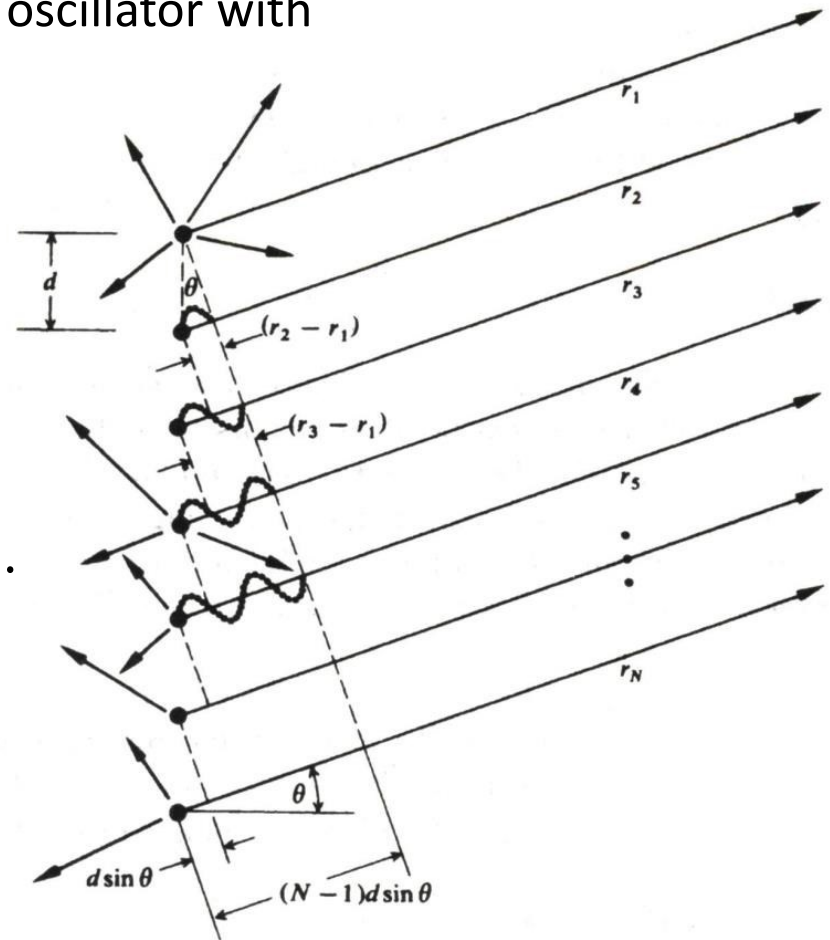
$$E_0(r_1) = E_0(r_2) = \dots = E_0(r_N) = E_0(r)$$

The sum of the interfering spherical wavelets at P

$$E = E_0(r)e^{i(kr_1 - \omega t)} + E_0(r)e^{i(kr_2 - \omega t)} + \dots + E_0(r)e^{i(kr_N - \omega t)}$$

$$E = E_0(r)e^{i(kR - \omega t)} \left(\frac{\sin N\delta/2}{\sin \delta/2} \right)$$

$$I = I_0 \left(\frac{\sin^2 N\delta/2}{\sin^2 \delta/2} \right)$$



$$E = E_0(r)e^{i(kr_1 - \omega t)} + E_0(r)e^{i(kr_2 - \omega t)} + \dots + E_0(r)e^{i(kr_N - \omega t)}$$

$$E = E_0(r)e^{-i\omega t}e^{ikr_1}[1 + e^{ik(r_2 - r_1)} + e^{ik(r_3 - r_1)} + \dots + e^{ik(r_N - r_1)}]$$

The phase difference between two adjacent point sources is $\delta = k_0\Lambda$

Considering the refractive index (n) of the medium $\Lambda = nd\sin\theta$

$$\delta = k_0\Lambda = kdsin\theta$$

$$E = E_0(r)e^{-i\omega t}e^{ikr_1}[1 + (e^{i\delta}) + (e^{i\delta})^2 + (e^{i\delta})^3 + \dots + (e^{i\delta})^{N-1}]$$

$$1 + (e^{i\delta}) + (e^{i\delta})^2 + (e^{i\delta})^3 + \dots + (e^{i\delta})^{N-1} = \frac{e^{i\delta N} - 1}{e^{i\delta} - 1} = \frac{e^{i\delta N/2}[e^{i\delta N/2} - e^{-i\delta N/2}]}{e^{i\delta/2}[e^{i\delta/2} - e^{-i\delta/2}]}$$

$$E = E_0(r)e^{-i\omega t}e^{i[kr_1 + (N-1)\delta/2]} \left(\frac{\sin N\delta/2}{\sin \delta/2} \right) = e^{i\delta(N-1)/2} \frac{\sin N\delta/2}{\sin \delta/2}$$

Define R as the distance from the center of the line of oscillators to the point-P

$$R = \frac{1}{2}(N-1)d\sin\theta + r_1$$

$$E = E_0(r)e^{i(kR - \omega t)} \left(\frac{\sin N\delta/2}{\sin \delta/2} \right) \quad I = I_0 \left(\frac{\sin^2 N\delta/2}{\sin^2 \delta/2} \right)$$

The flux-density distribution:

$$I = I_0 \left(\frac{\sin^2 N\delta/2}{\sin^2 \delta/2} \right) = I_0 \frac{\sin^2 \left[N \left(\frac{kd}{2} \right) \sin \theta \right]}{\sin^2 \left[\left(\frac{kd}{2} \right) \sin \theta \right]}$$

I_0 : flux density from any single source arriving at P

(I) The principal maxima: $N^2 I_0$.

$$\sin N\delta/2 \neq 0 \quad \text{and} \quad \sin \delta/2 = 0$$

$$\delta = 2m\pi, m = 0, \pm 1, \pm 2, \dots$$

(II) The minimum: zero

$$\sin N\delta/2 = 0 \quad \text{but} \quad \sin \delta/2 \neq 0$$

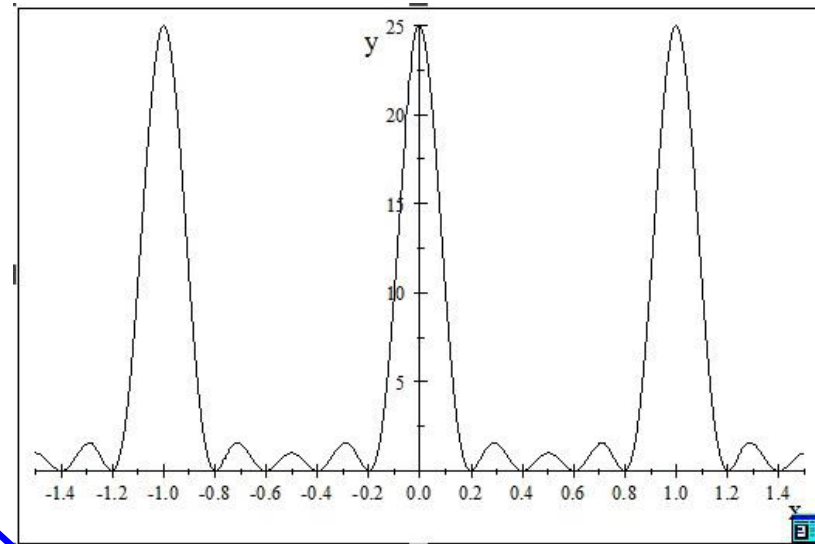
$$\frac{\delta}{2} = \pm \frac{\pi}{N}, \pm \frac{2\pi}{N}, \pm \frac{3\pi}{N}, \dots, \pm \frac{(N-1)\pi}{N}, \pm \frac{(N+1)\pi}{N}$$

(III) The (N-2) subsidiary maxima

$$\sin N\delta/2 = \pm 1$$

$$\frac{\delta}{2} = \pm \frac{\pi}{2N}, \pm \frac{3\pi}{2N}, \pm \frac{5\pi}{2N}, \dots$$

The expression gives rise to a series of sharp principal peaks separated by small subsidiary maxima.



L'Hospital's Rule
(洛必达法则)

When $\delta = 2m\pi$

$$\frac{\sin^2 N\delta/2}{\sin^2 \delta/2} = N^2$$

The idealized line source of electron-oscillator: the sources are very weak, their number (N) is tremendously large, and separation between them is vanishingly small.

Each point emits a spherical wavelet:

$$E = \left(\frac{\epsilon_0}{r} \right) \sin(\omega t - kr)$$

ϵ_0 : the source strength (波源强度).

Source number in a minute but finite segment Δy_i (the oscillators within Δy_i have a negligible relative phase difference):

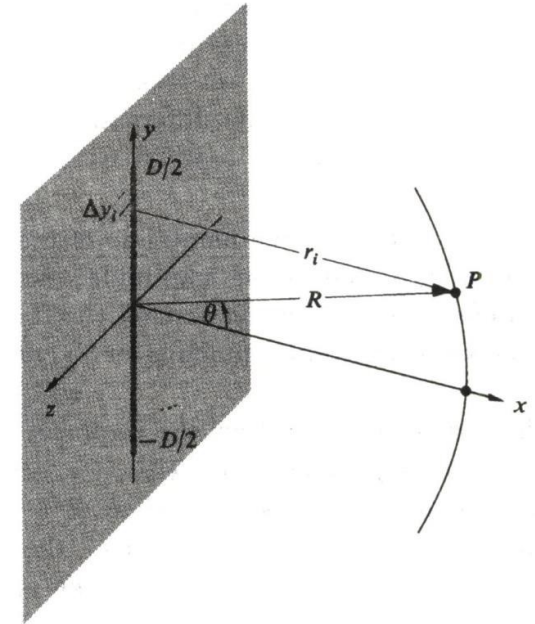
$$\frac{N\Delta y_i}{D} \quad \begin{array}{l} N \text{ is the total number of the electron-oscillators;} \\ D \text{ is the entire length of the line source.} \end{array}$$

Total M such segments, the contribution to electric-field intensity at P from the i -th segment:

$$E_i = \left(\frac{\epsilon_0}{r_i} \right) \sin(\omega t - kr_i) \left(\frac{N\Delta y_i}{D} \right)$$

Source strength per unit length for infinite N .

$$\epsilon_L \equiv \frac{1}{D} \lim_{N \rightarrow \infty} (\epsilon_0 N)$$



The net field at P :

$$E_i = \sum_{i=1}^M \left(\frac{\epsilon_L}{r_i} \right) \sin(\omega t - kr_i) \Delta y_i$$

For a continuous line source:

$$E_i = \epsilon_L \int_{-D/2}^{D/2} \frac{\sin(\omega t - kr_i)}{r} dy$$

Condition: the point of observation is very distant from the coherent line source and $R \gg D$: $r(y) \approx R$

$$dE = \frac{\epsilon_L}{R} \sin(\omega t - kr) dy$$

Expand $r(y)$:

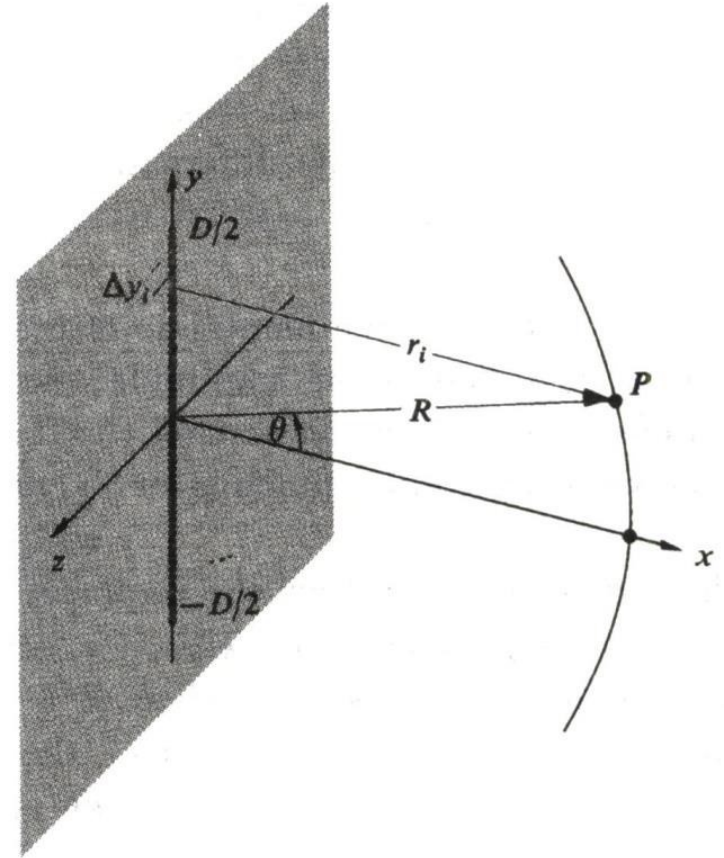
$$r = R - y \sin \theta + \left(\frac{y^2}{2R} \right) \cos^2 \theta + \dots$$

Fraunhofer condition: R is large enough

$$r = R - y \sin \theta$$

$$E = \frac{\epsilon_L}{R} \int_{-D/2}^{D/2} \sin[\omega t - k(R - y \sin \theta)] dy$$

$$E = \frac{\epsilon_L D}{R} \left(\frac{\sin \beta}{\beta} \right) \sin(\omega t - kR) \quad \beta \equiv \left(\frac{kD}{2} \right) \sin \theta$$



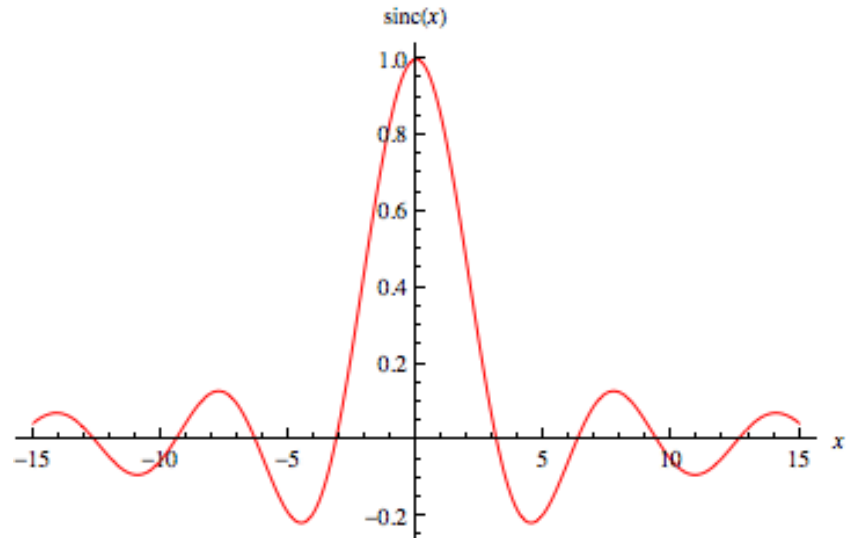
Irradiance: $I(\theta) = \langle E^2 \rangle_T$

$$I(\theta) = \left(\frac{\epsilon_L D}{R} \right)^2 \left(\frac{\sin \beta}{\beta} \right)^2 \langle \sin^2(\omega t - kR) \rangle_T = \frac{1}{2} \left(\frac{\epsilon_L D}{R} \right)^2 \left(\frac{\sin \beta}{\beta} \right)^2$$

$\theta = 0, \frac{\sin \beta}{\beta} = 1$ and $I(\theta) = I(0)$: the principal maximum

$$I(\theta) = I(0) \left(\frac{\sin \beta}{\beta} \right)^2$$

$$\beta \equiv \left(\frac{kD}{2} \right) \sin \theta = \left(\frac{\pi D}{\lambda} \right) \sin \theta$$



(I) $D \gg \lambda$:

The irradiance drops extremely rapidly as θ deviates from zero. So the relatively long coherent line source can be envisioned as a single-point emitter radiating in the forward direction ($\theta=0$).

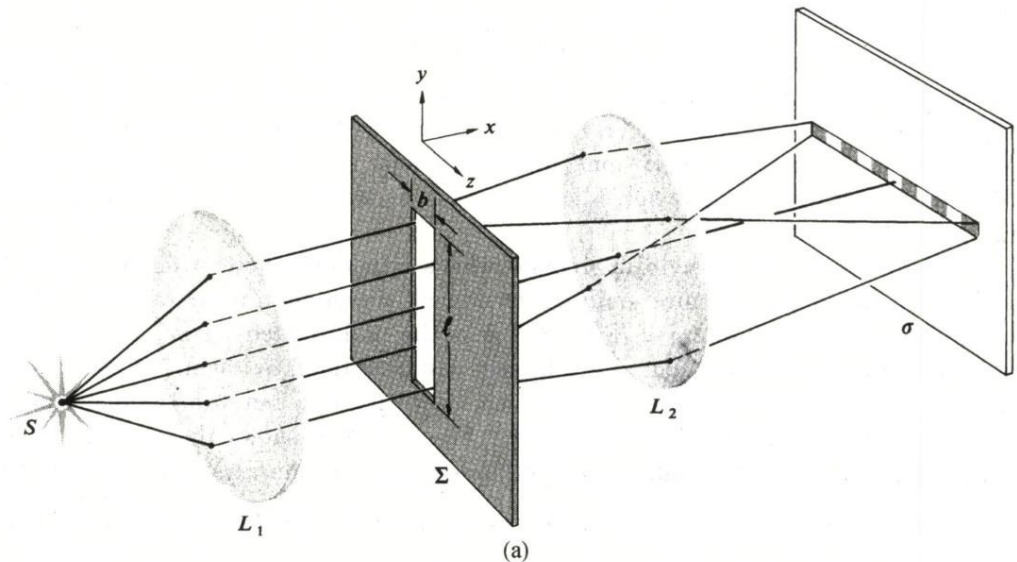
(II) $D \ll \lambda$: $\sin \beta \approx \beta, I(\theta) = I(0)$

The line source resembles a point source emitting spherical waves.

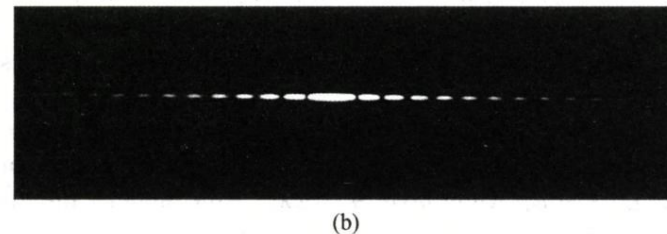
9.2 Fraunhofer Diffraction

The single slit Fraunhofer diffraction by a slit or elongated, narrow rectangular hole.

- ❑ Width: several hundred λ .
- ❑ Length: a few centimeters.



Each strip is a long coherent line source and can be replaced by a point emitter on the z -axis, which radiates a circular wave in the xz -plane.



The problem is reduced to that of finding the field in the xz -plane, once again, this is equivalent to a coherent line source.

$$I(\theta) = I(0) \left(\frac{\sin \beta}{\beta} \right)^2$$

$$\beta \equiv \left(\frac{kb}{2} \right) \sin \theta$$

The extrema of $I(\theta)$ occur at $dI/d\beta=0$

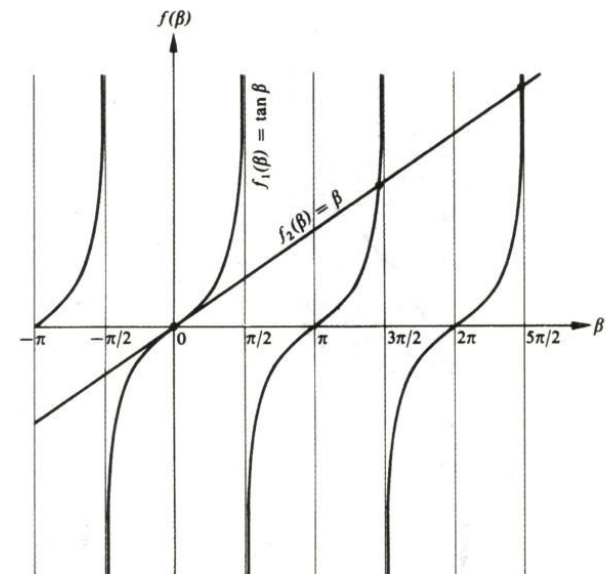
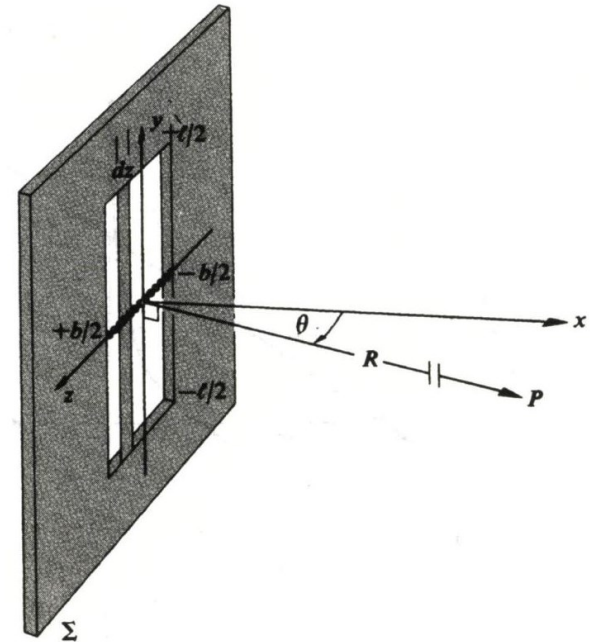
$$\frac{dI}{d\beta} = I(0) \frac{2\sin \beta (\beta \cos \beta - \sin \beta)}{\beta^3} = 0$$

□ The minima of irradiance: zero ($\sin \beta=0$):

$$\beta = \pm\pi, \pm2\pi, \pm3\pi, \dots$$

□ The subsidiary maxima: $\beta \cos \beta - \sin \beta = 0 \Rightarrow \tan \beta = \beta$

$$\beta = \pm 1.43\pi, \pm 2.46\pi, \pm 3.47\pi, \dots$$



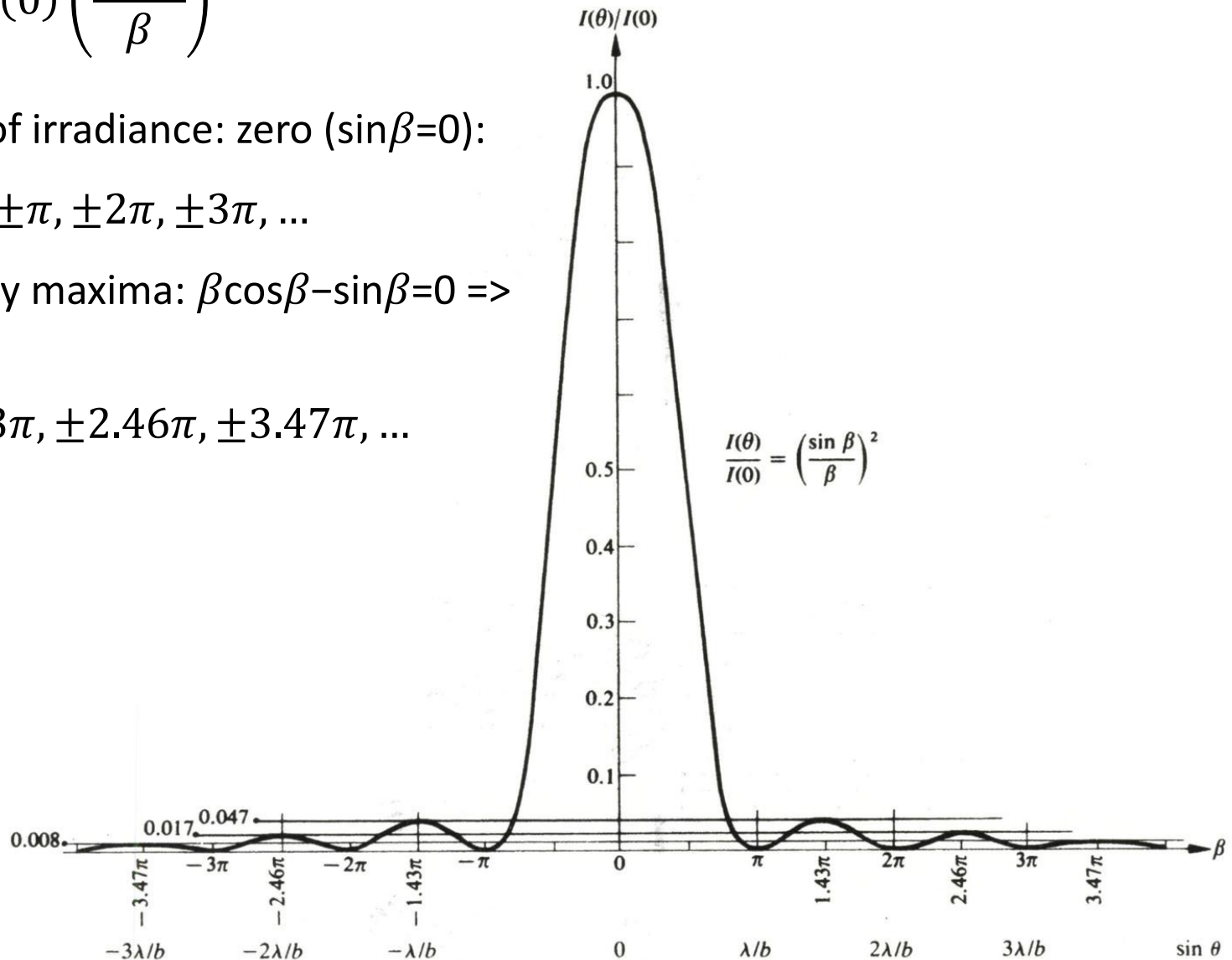
$$I(\theta) = I(0) \left(\frac{\sin \beta}{\beta} \right)^2$$

□ The minima of irradiance: zero ($\sin \beta = 0$):

$$\beta = \pm\pi, \pm2\pi, \pm3\pi, \dots$$

□ The subsidiary maxima: $\beta \cos \beta - \sin \beta = 0 \Rightarrow \tan \beta = \beta$

$$\beta = \pm 1.43\pi, \pm 2.46\pi, \pm 3.47\pi, \dots$$



The light propagates directly forward:

- ❑ Undiffracted beam; In phase; The central bright spot.

The path length difference between the rays from top and bottom is one wavelength:

- ❑ The irradiance drops to zero;

$$\sin \theta_1 = \pm \lambda / b$$

The path difference equals to two wavelength:

- ❑ The irradiance drops to zero $I(\theta_2) = 0$;

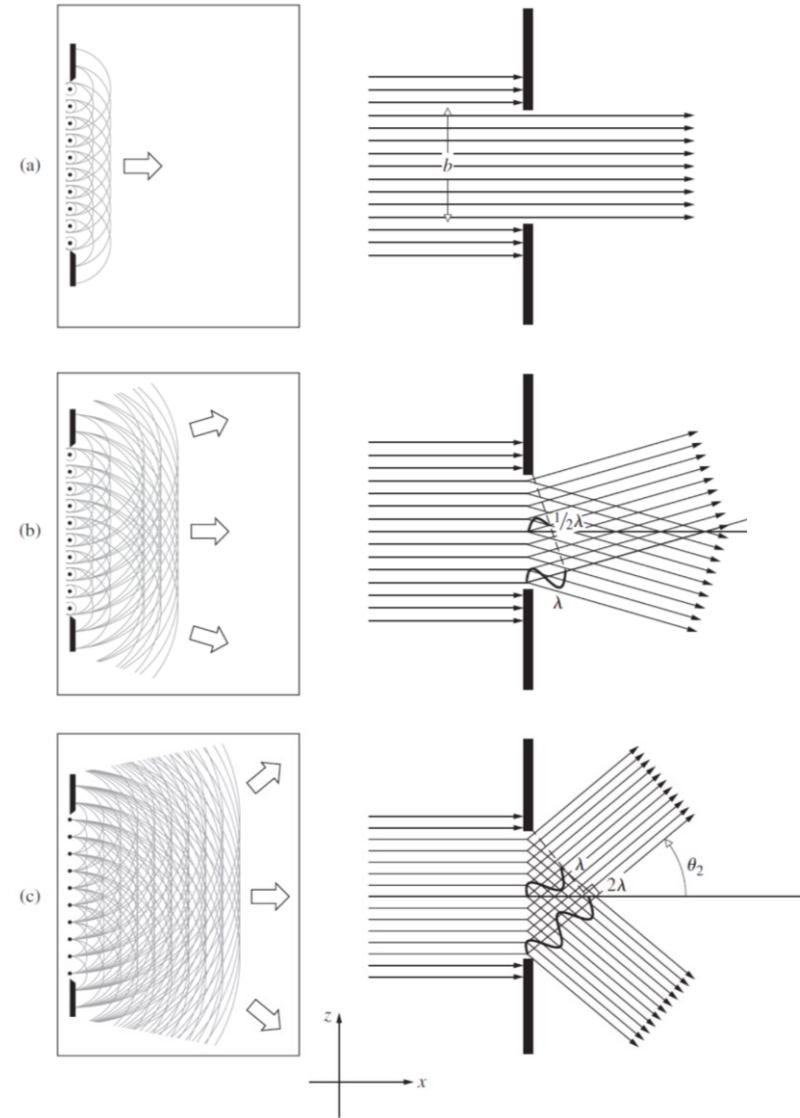
$$\sin \theta_2 = \pm 2 \lambda / b$$

In general:

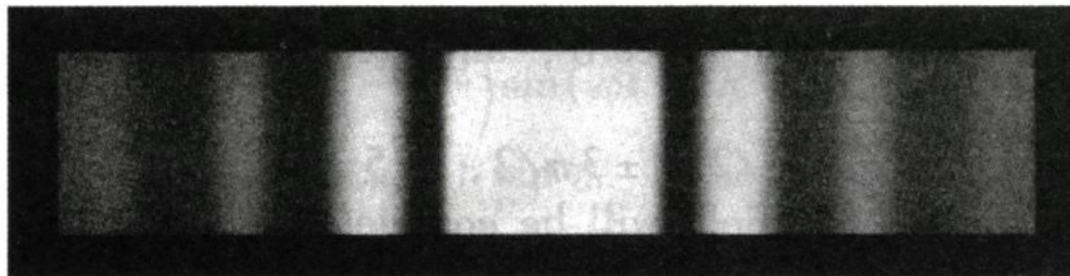
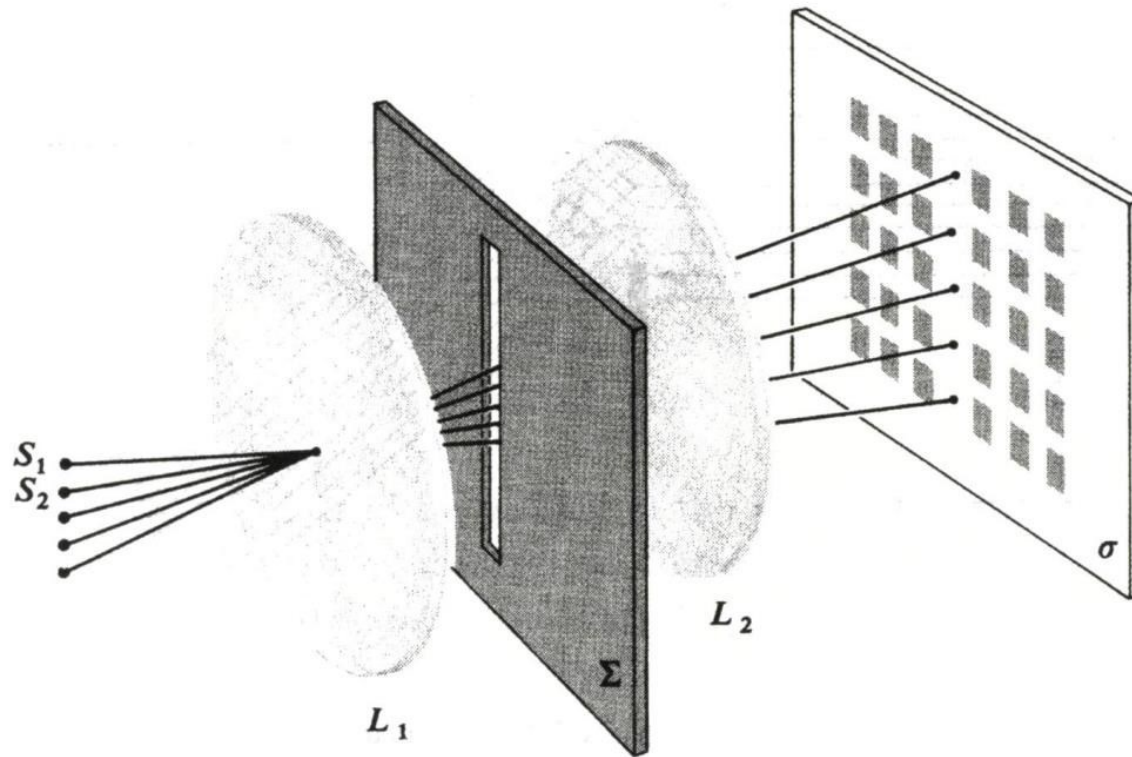
$$b \sin \theta_m = m \lambda$$

$$m = \pm 1, \pm 2, \pm 3, \dots$$

which is equal to minimum discussed above.

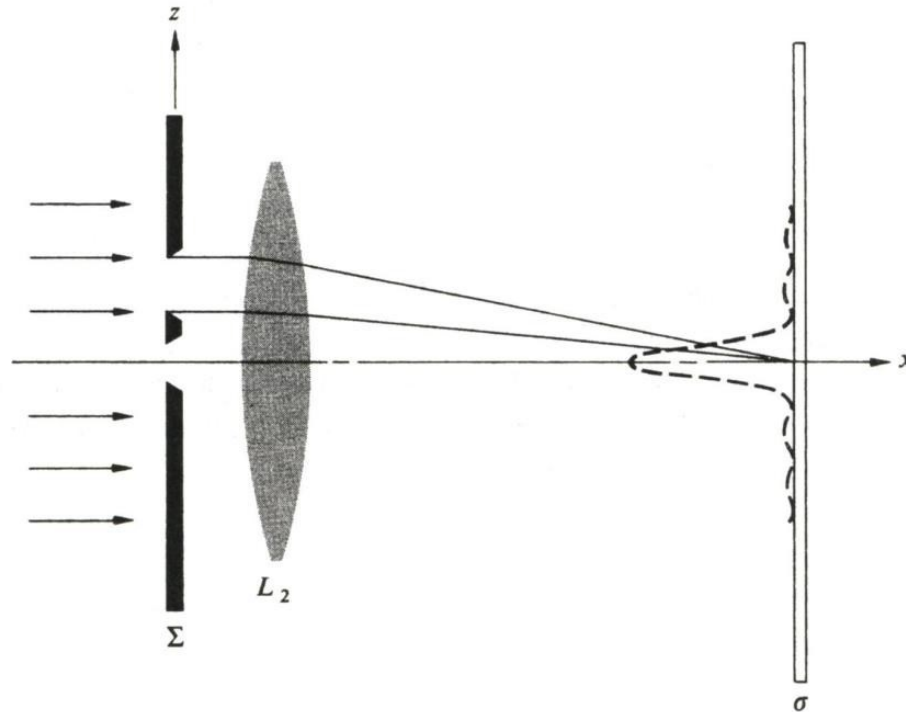


An incoherent line source positioned parallel to the slit.



The Double Slit

The diffraction pattern is centered about the axis of the lens, regardless of the slit's position



The double slit diffraction:

- ☐ Each aperture generate single-slit diffraction pattern.
- ☐ Two slits will interfere, which is determined by the differences of optical path lengths traversed by the overlapping wavelets from the two slits.

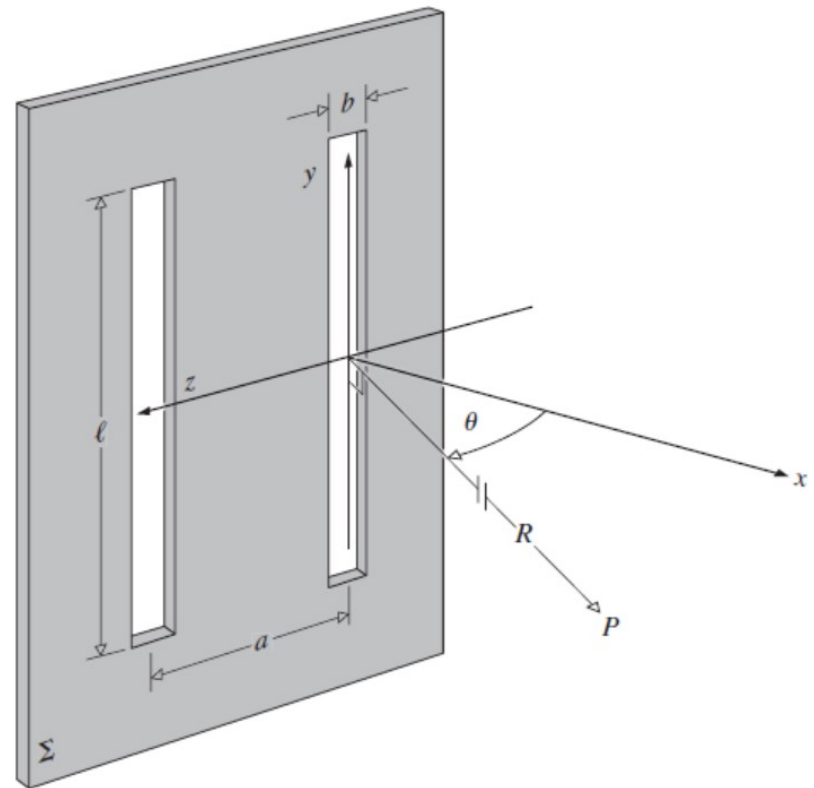
The expression of optical disturbance at a point on screen for double slit diffraction:

$$E = C \int_{-\frac{b}{2}}^{\frac{b}{2}} F(z) dz + C \int_{a-\frac{b}{2}}^{a+\frac{b}{2}} F(z) dz$$

$$F(z) = \sin[\omega t - k(R - z \sin \theta)]$$

$$E = bC \left(\frac{\sin \beta}{\beta} \right) [\sin(\omega t - kR) + \sin(\omega t - kR + 2\alpha)]$$

$$\beta \equiv \left(\frac{kb}{2} \right) \sin \theta \quad \alpha \equiv \left(\frac{ka}{2} \right) \sin \theta$$



This is just the sum of the two fields at P, one from each slit.

$$E = 2bC \left(\frac{\sin \beta}{\beta} \right) \cos \alpha \sin(\omega t - kR + \alpha)$$

$$I(\theta) = 2b^2 C^2 \left(\frac{\sin^2 \beta}{\beta^2} \right) \cos^2 \alpha = 4I_0 \left(\frac{\sin^2 \beta}{\beta^2} \right) \cos^2 \alpha$$

$$\beta \equiv \left(\frac{kb}{2} \right) \sin \theta \quad \alpha \equiv \left(\frac{ka}{2} \right) \sin \theta$$

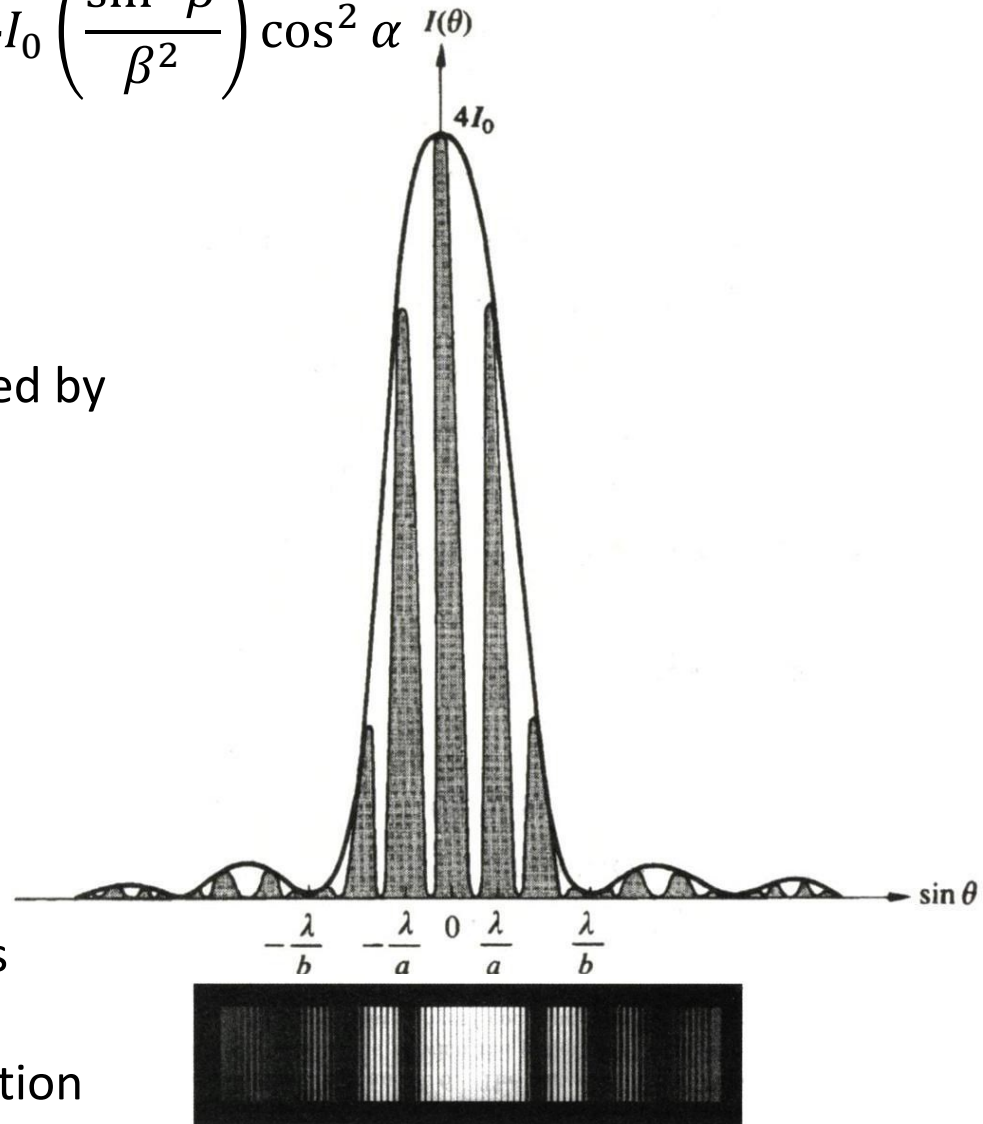
A $\cos^2 \alpha$ interference term modulated by a $\sin^2 \beta / \beta^2$ diffraction term.

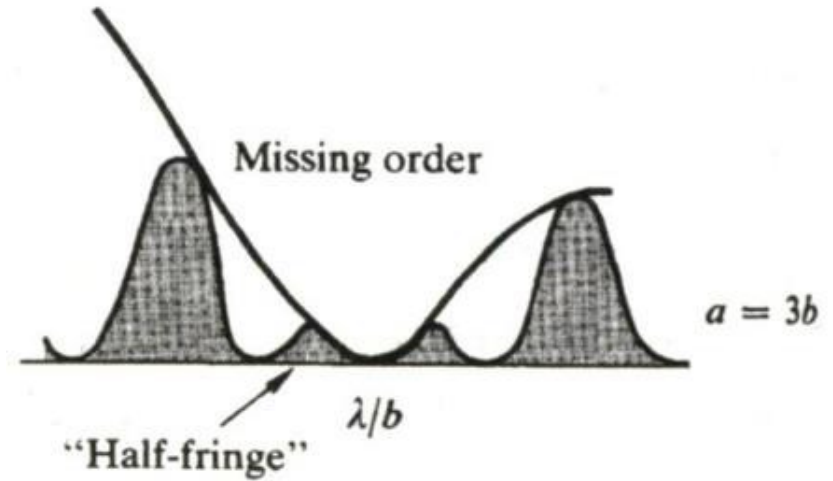
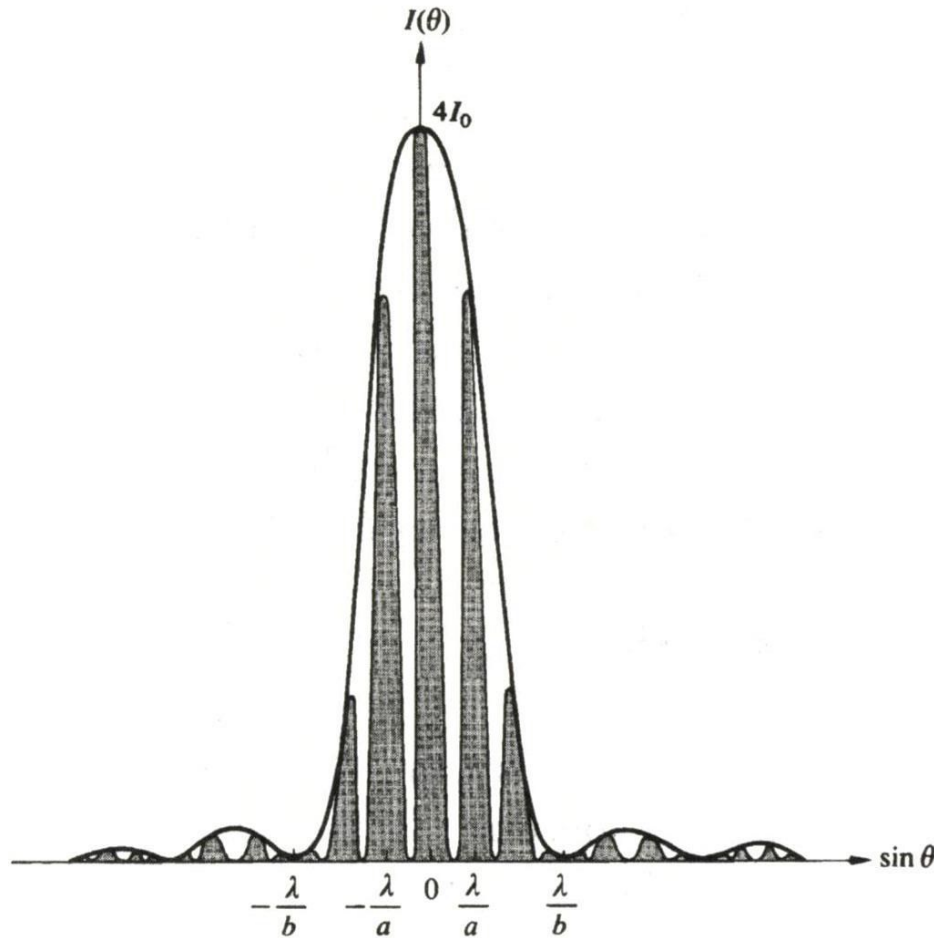
$I(\theta) = 0$, when

$$\beta = \pm\pi, \pm2\pi, \pm3\pi, \dots$$

$$\alpha = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \pm\frac{5\pi}{2}, \dots$$

- (I) $kb \ll 1$, then $(\sin \beta)/\beta \approx 1$: Young's experiment
- (II) $a=0$, then $\alpha=0$: single-slit diffraction





Missing order: an interference maximum and a diffraction minimum occur at the same θ -value.

Example: when $a=mb$

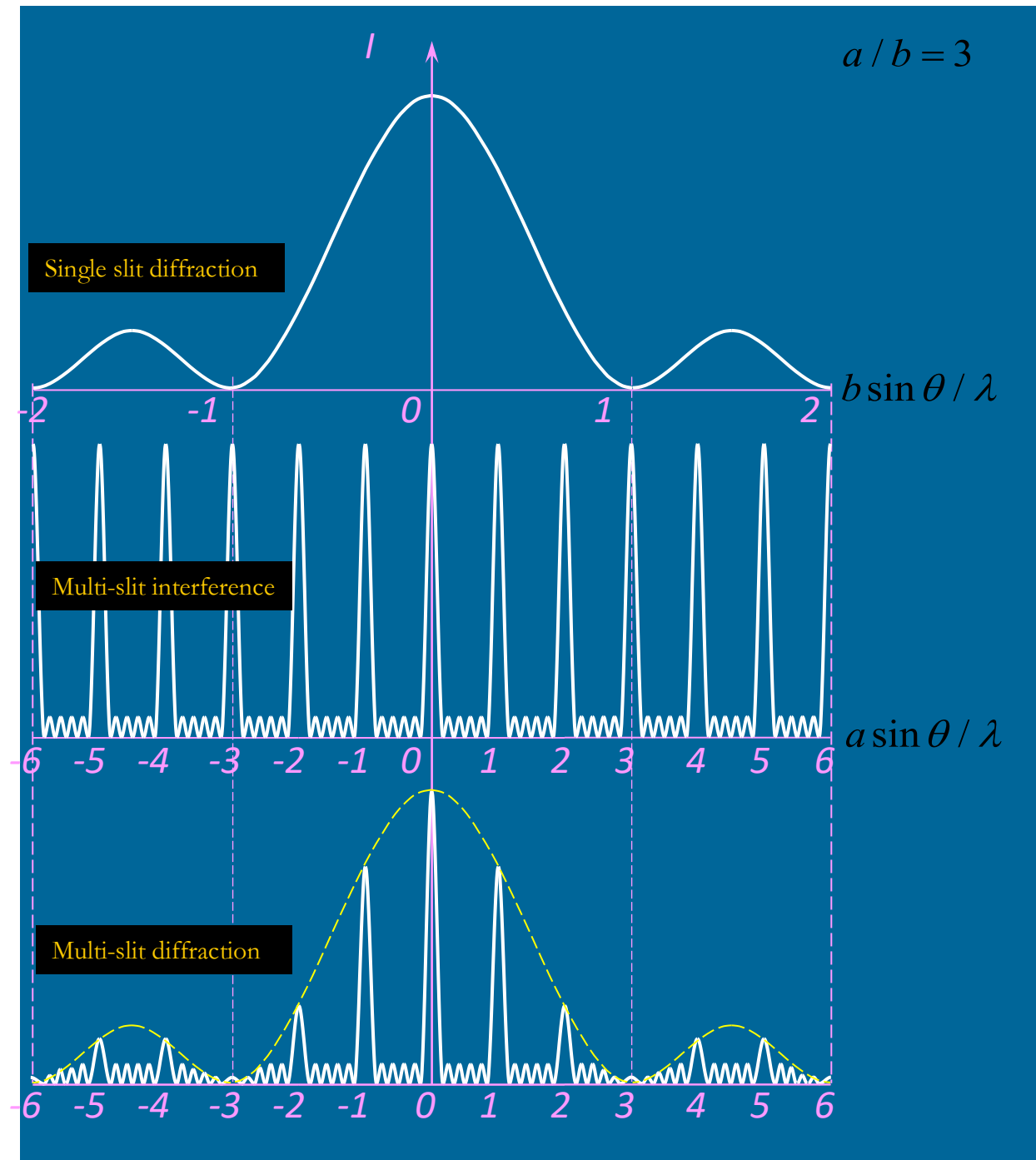
$$\text{Number of fringes: } N = \frac{\frac{2\lambda}{b}}{\frac{\lambda}{a}} = \frac{2a}{b} = 2m$$

There is $2m$ bright fringes within the central diffraction peak.

$$I(\theta) = 4I_0 \left(\frac{\sin^2 \beta}{\beta^2} \right) \cos^2 \alpha$$

$$\beta \equiv \left(\frac{kb}{2} \right) \sin \theta$$

$$\alpha \equiv \left(\frac{ka}{2} \right) \sin \theta$$



Diffraction by Many Slits

Diffraction by N slits with the width b and center-to-center separation a .

$$\begin{aligned}
 E = & C \int_{-\frac{b}{2}}^{\frac{b}{2}} F(z) dz + C \int_{a-\frac{b}{2}}^{a+\frac{b}{2}} F(z) dz \\
 & + C \int_{2a-\frac{b}{2}}^{2a+\frac{b}{2}} F(z) dz + \dots \\
 & + C \int_{(N-1)a-\frac{b}{2}}^{(N-1)a+\frac{b}{2}} F(z) dz
 \end{aligned}$$

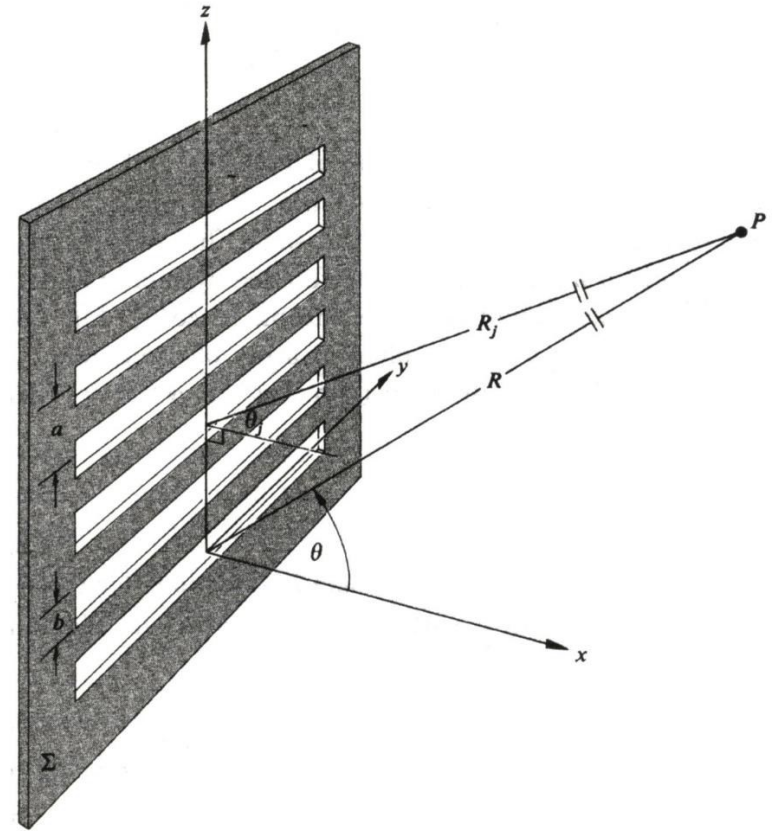
$$F(z) = \sin[\omega t - k(R - z \sin \theta)]$$

Fraunhofer condition: $r = R - z \sin \theta$

Require $\theta_j \approx \theta$

$$E = bC \left(\frac{\sin \beta}{\beta} \right) \left(\frac{\sin N\alpha}{\sin \alpha} \right) \sin[\omega t - kR + (N-1)\alpha]$$

$R - (N-1) \left(\frac{a}{2} \right) \sin \theta$: the distance from the center of the array to the point P.



The flux-density distribution function:

$$I(\theta) = I_0 \left(\frac{\sin \beta}{\beta} \right)^2 \left(\frac{\sin N\alpha}{\sin \alpha} \right)^2$$

I_0 is the flux density on the $\theta = 0$ direction emitted by any one of the slits; $I(0) = N^2 I_0$.

A multiple-wave interference

$\left(\frac{\sin N\alpha}{\sin \alpha} \right)^2$ system modulated by the single-slit diffraction envelop

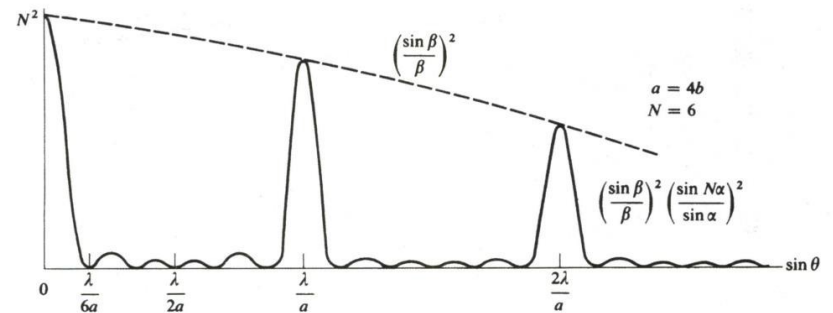
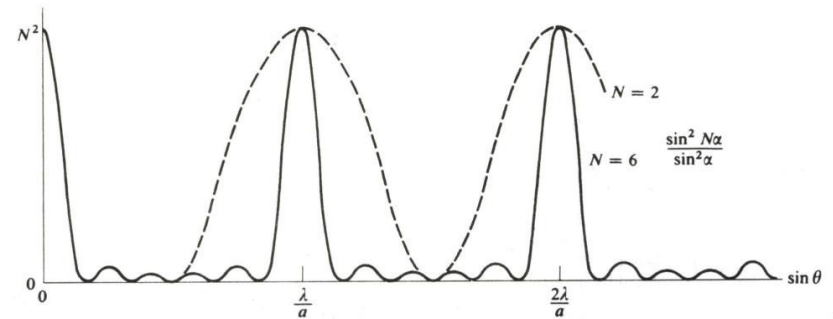
$$\left(\frac{\sin \beta}{\beta} \right)^2$$

$$N = 1, \quad I(\theta) = I_0 \left(\frac{\sin \beta}{\beta} \right)^2 \left(\frac{\sin \alpha}{\sin \alpha} \right)^2 = I_0 \left(\frac{\sin \beta}{\beta} \right)^2$$

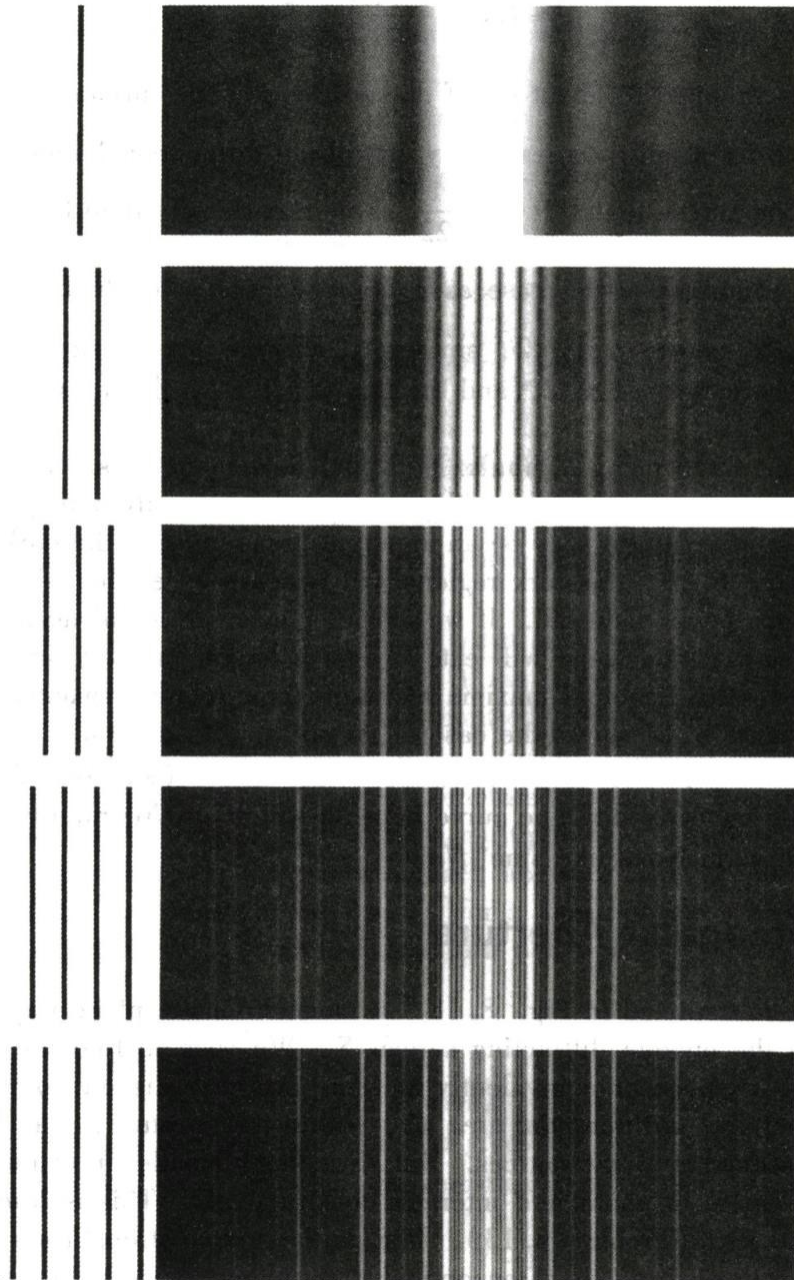
single slit diffraction formula

$$N = 2, \quad I(\theta) = I_0 \left(\frac{\sin \beta}{\beta} \right)^2 \left(\frac{\sin 2\alpha}{\sin \alpha} \right)^2 = 4I_0 \left(\frac{\sin \beta}{\beta} \right)^2 \cos^2 \alpha$$

double slit diffraction formula



Between two adjacent principle irradiance maxima, there are $(N-1)$ minima and $(N-2)$ subsidiary maxima.



The flux-density distribution function:

$$I(\theta) = I_0 \left(\frac{\sin \beta}{\beta} \right)^2 \left(\frac{\sin N\alpha}{\sin \alpha} \right)^2$$

Between two adjacent principle irradiance maxima, there are $(N-1)$ minima and $(N-2)$ subsidiary maxima.

Homework

□ 10.2, 10.10, 10.12

The Diffraction Grating

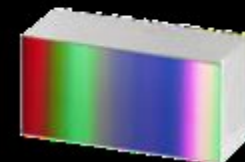
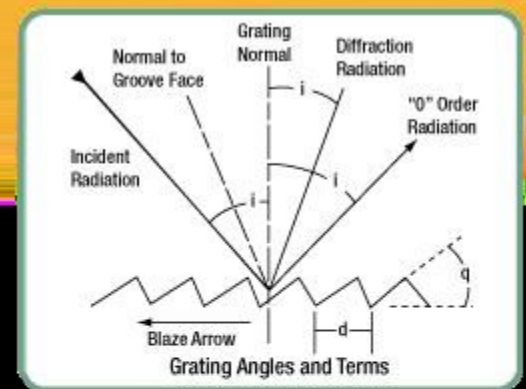
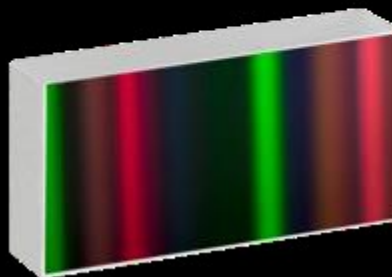
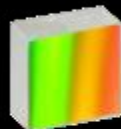
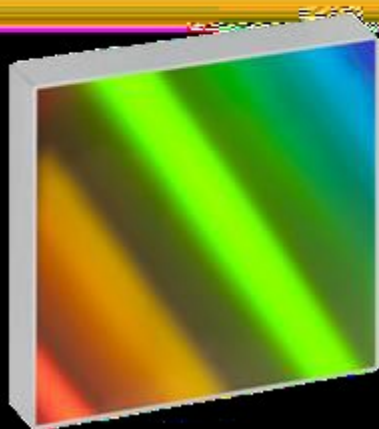
A repetitive array of diffracting elements, either apertures or obstacles, that has the effect of producing periodic alterations in the phase, amplitude, or both of an emergent wave.

Reflection phase grating

300 nm to 10.6 μm Blaze Wavelengths

Higher Efficiencies than Holographic Gratings

5 Sizes Available with Sides Measuring from 1/2" to 2"



Recall: **diffraction by many slits**

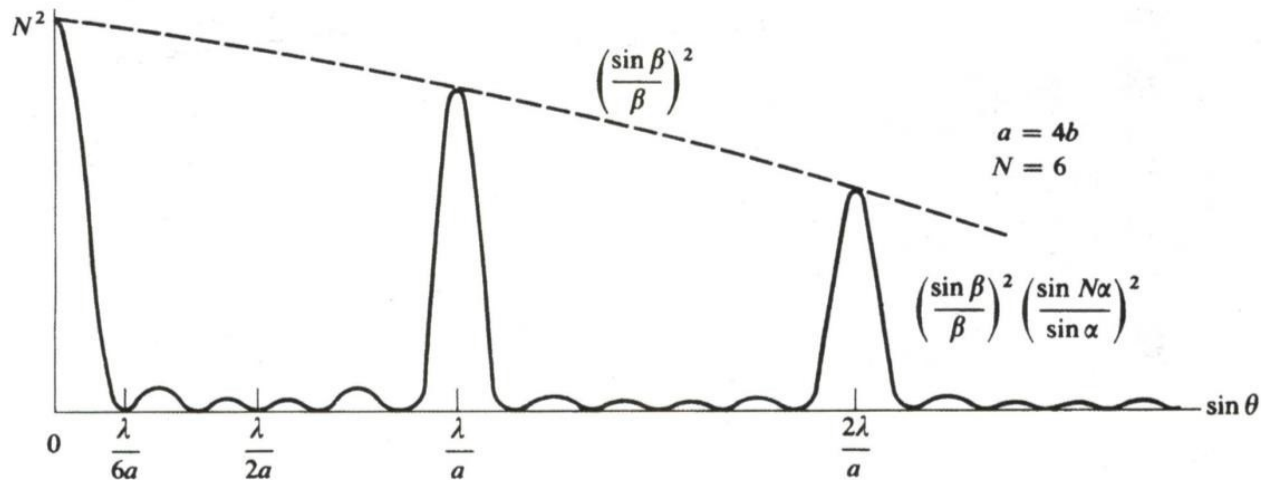
Irradiance at P:
$$I(\theta) = I_0 \left(\frac{\sin \beta}{\beta} \right)^2 \left(\frac{\sin N\alpha}{\sin \alpha} \right)^2$$

$$\beta \equiv \left(\frac{kb}{2} \right) \sin \theta \quad \alpha \equiv \left(\frac{ka}{2} \right) \sin \theta$$

Principal maxima: $\sin N\alpha = 0$ and $\sin \alpha = 0$

$$\alpha = 0, \pm\pi, \pm2\pi, \dots \pm m\pi$$

$$a \sin \theta_m = m\lambda$$

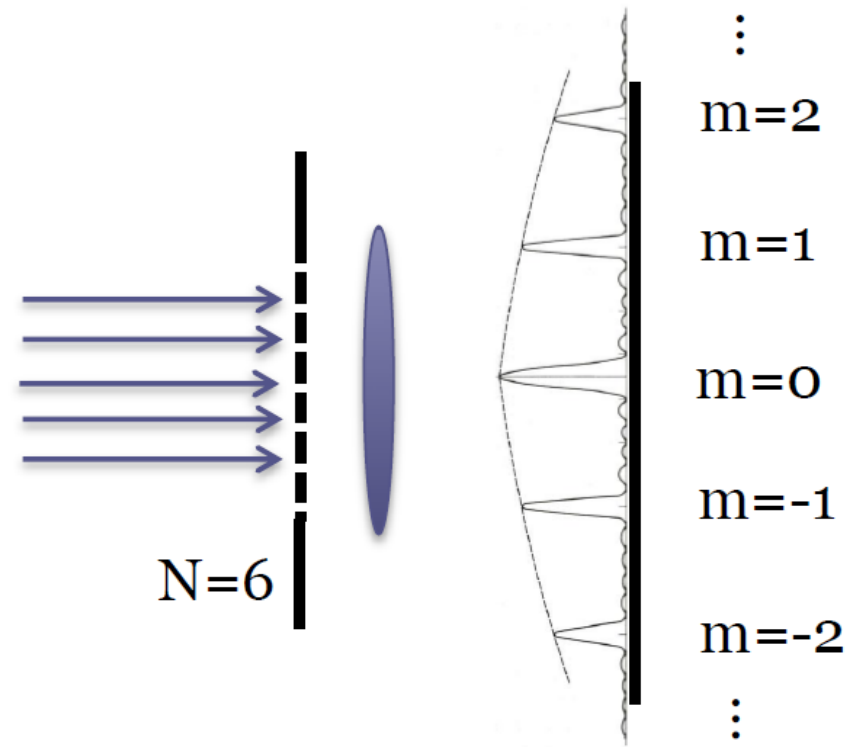
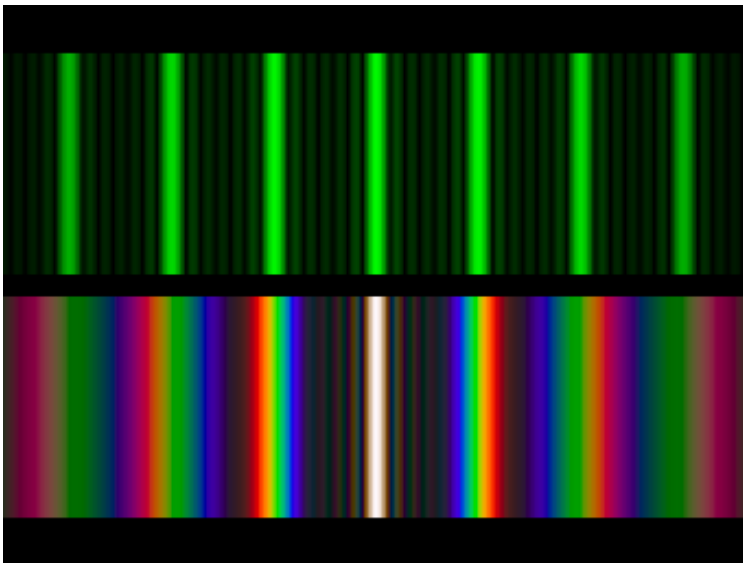


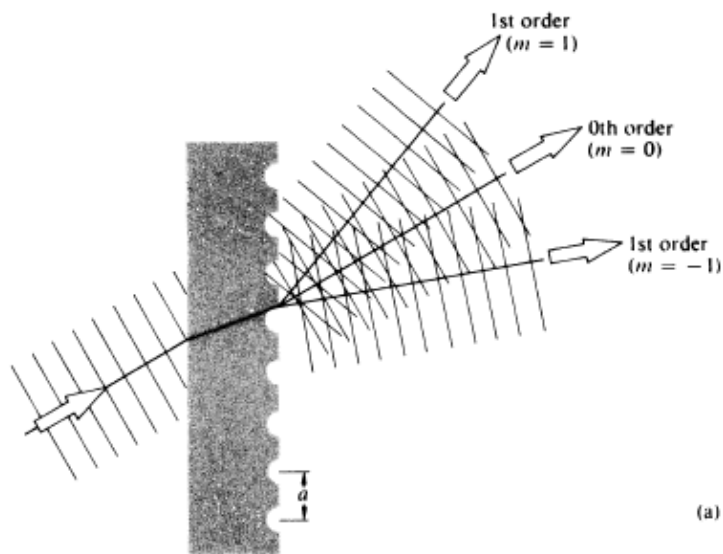
Grating equation for normal incidence:

$$a \sin \theta_m = m\lambda$$

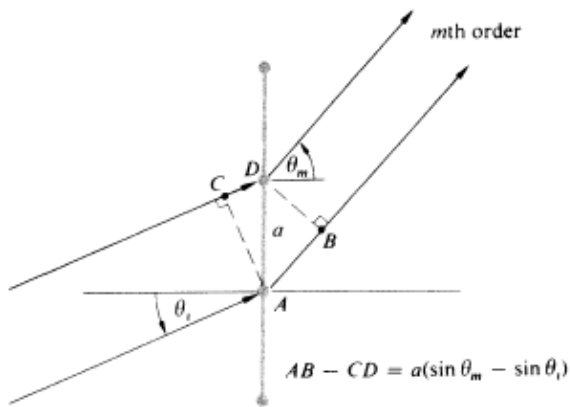
For $m=0$, $\theta_0=0$: image corresponds to the undeflected, white-light view of the source.

For $m \neq 0$, the grating equation is dependent on λ : the various colored images of the source corresponding to slightly different angles.

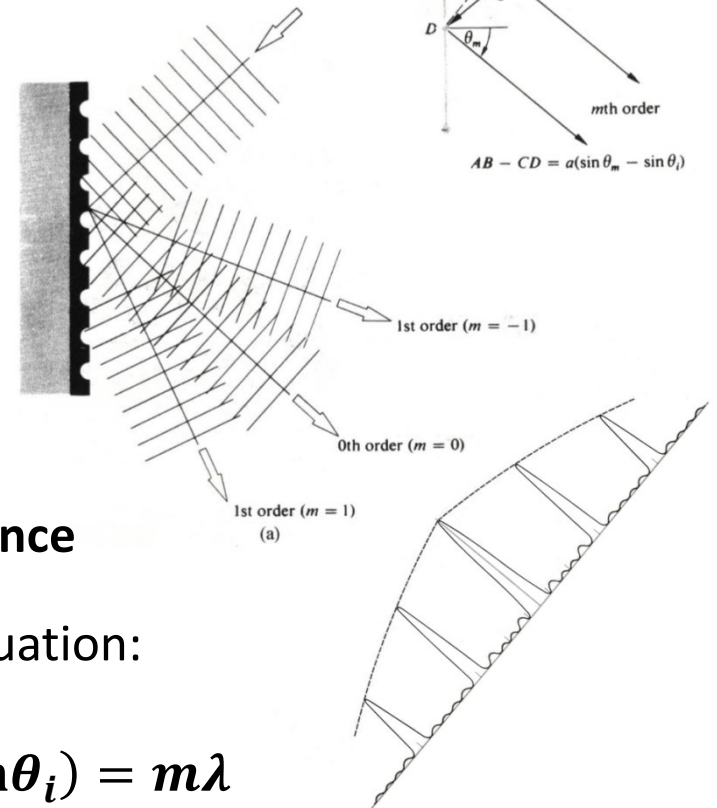




(a)



(b)



Oblique incidence

The grating equation:

$$a(\sin\theta_m - \sin\theta_i) = m\lambda$$

The zeroth order: $\theta_m = \theta_i$.

For spectroscopic purposes, all of the light is essentially wasted.

Idea: shift energy out of the useless zeroth order into one of the higher-order spectra.

Blazed reflection phase grating:

Recall

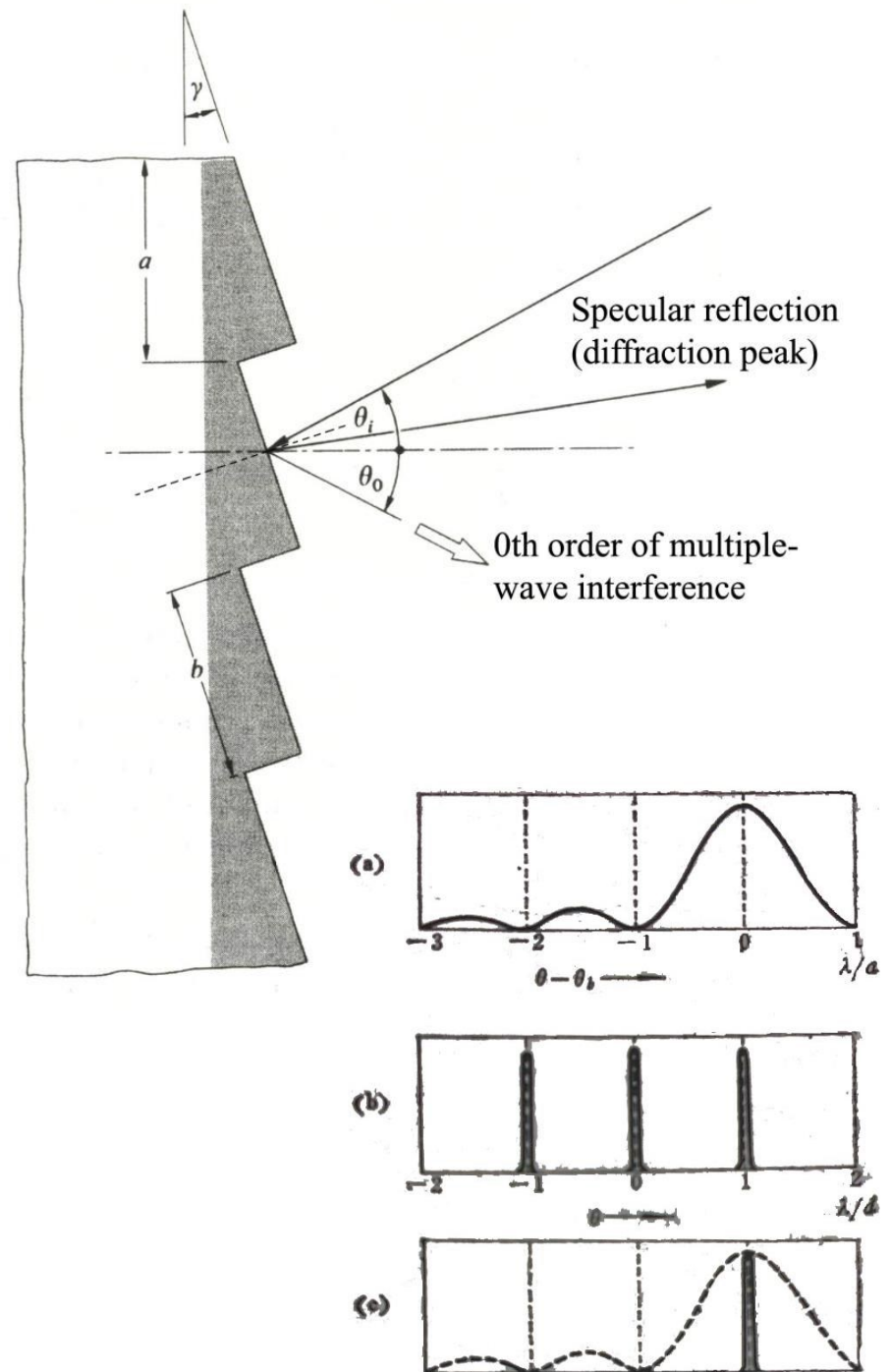
$$I(\theta) = I_0 \left(\frac{\sin \beta}{\beta} \right)^2 \left(\frac{\sin N\alpha}{\sin \alpha} \right)^2$$

Single-facet diffraction

$\left(\frac{\sin \beta}{\beta} \right)^2$ corresponds to specular reflection off that face, for each groove, which is governed by the blaze angle γ .

A multiple-wave interference

$\left(\frac{\sin N\alpha}{\sin \alpha} \right)^2$ or θ_m value are determined by a , λ and θ_i . But θ_m and θ_i are measured from the normal to the grating plane and not with respect to the individual groove surfaces.



Example: the normal incidence

Single-facet diffraction

$$\left(\frac{\sin \beta}{\beta}\right)^2 :$$

$$\theta_i - \theta_r = 2\gamma \rightarrow \theta_r = -2\gamma$$

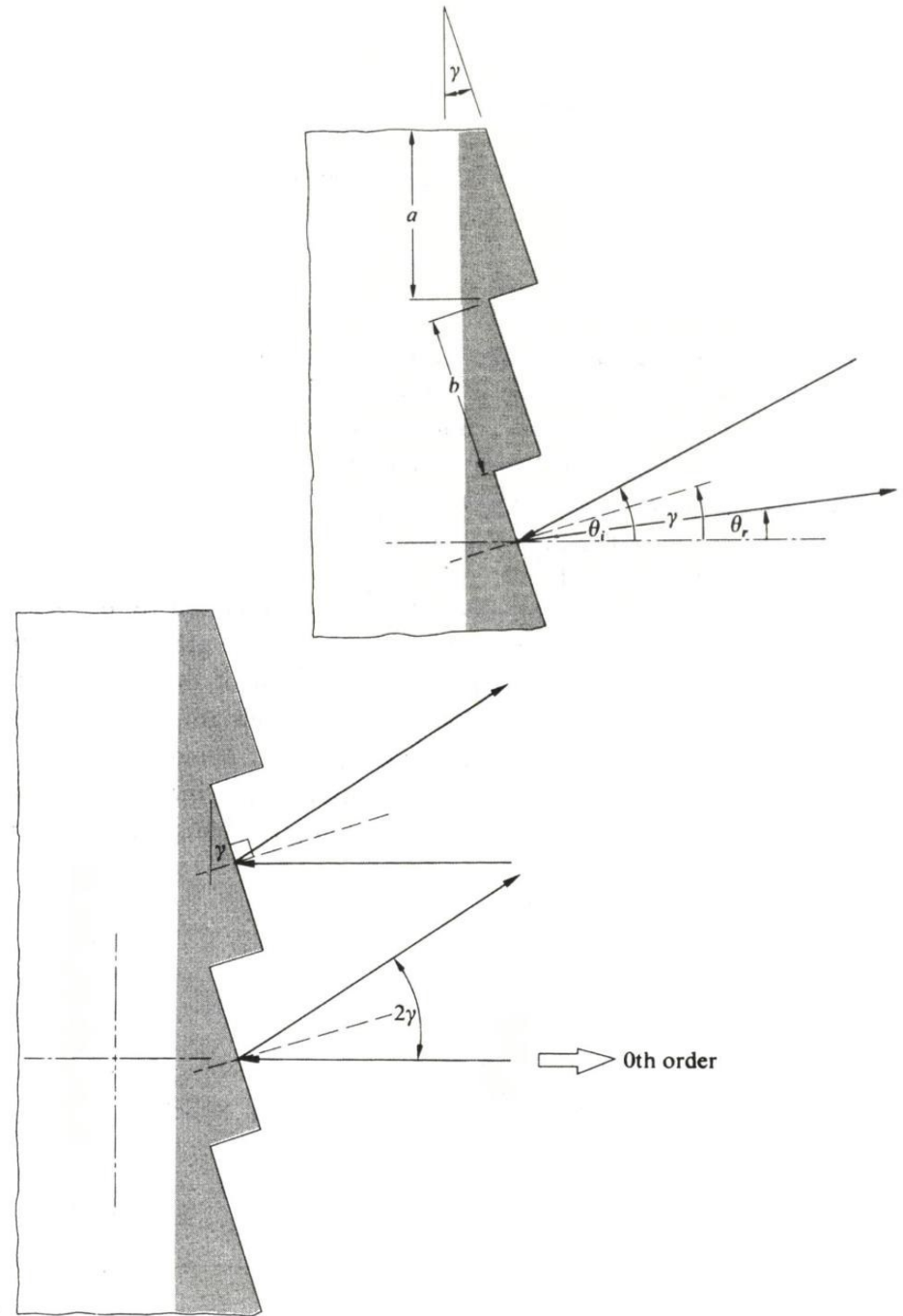
A multiple-wave interference term

$$\left(\frac{\sin N\alpha}{\sin \alpha}\right)^2 : \theta_i = 0$$

$$\text{For } m = 0 ; \theta_0 = 0$$

Principle maxima:

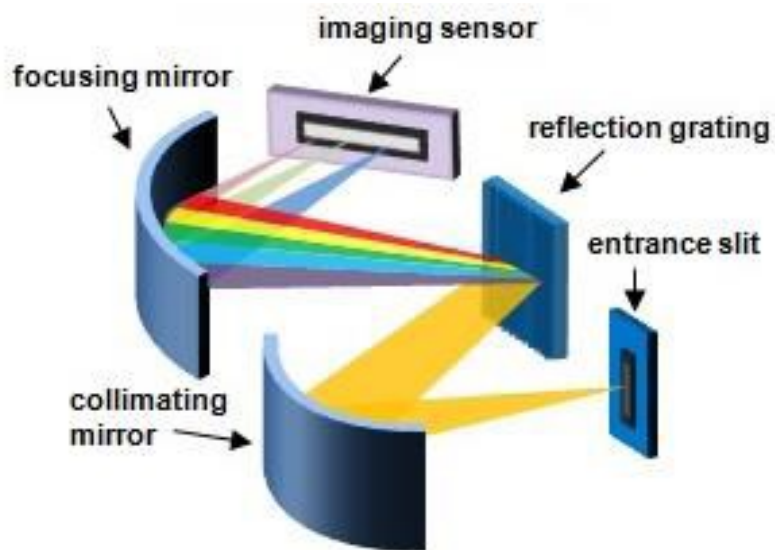
$$a \sin(-2\gamma) = m\lambda$$





Spectrometer

分光计



The Arbitrary Aperture

Huygens-Fresnel Principle:

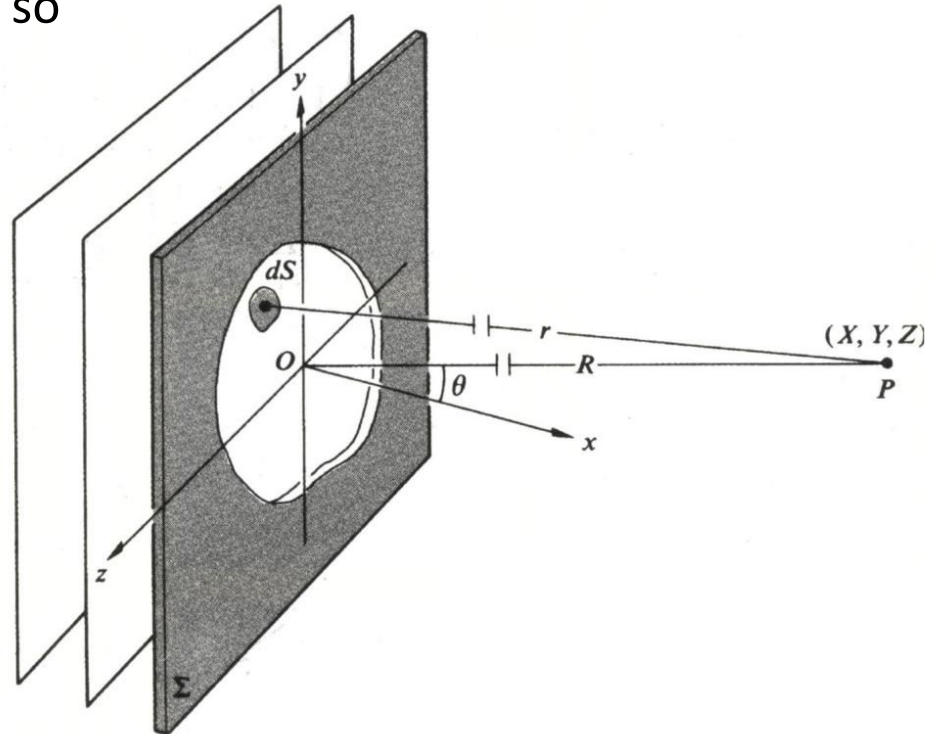
A differential area dS : it is very small, so that *all the contributions at P* are in phase; it emits a spherical wave.

The optical disturbance at P:

$$dE = \left(\frac{\epsilon_A}{R} \right) e^{i(\omega t - kr)} dS$$

Fraunhofer condition, the total disturbance:

$$\tilde{E} = \frac{\epsilon_A e^{i(\omega t - kR)}}{R} \int_{\text{Aperture}} e^{ik(Yy + Zz)} dS$$

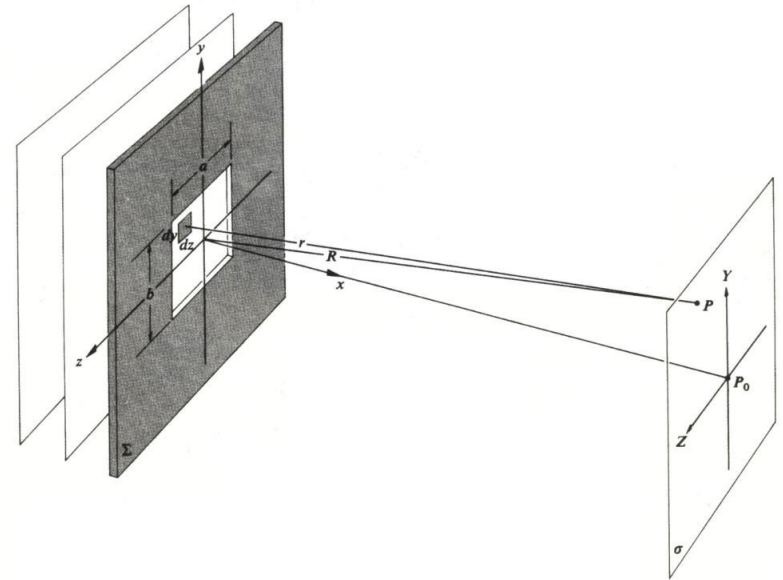


The rectangular aperture

$$\tilde{E} = \frac{\epsilon_A e^{i(\omega t - kR)}}{R} \int_{-\frac{b}{2}}^{\frac{b}{2}} e^{ikYy} dy \int_{-\frac{a}{2}}^{\frac{a}{2}} e^{ikZz} dz$$

$$\tilde{E} = \frac{A\epsilon_A e^{i(\omega t - kR)}}{R} \left(\frac{\sin \alpha'}{\alpha'} \right) \left(\frac{\sin \beta'}{\beta'} \right)$$

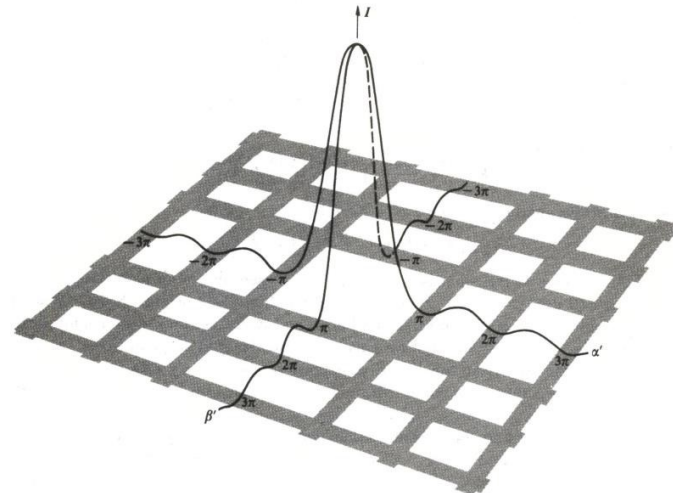
$$\alpha' \equiv kaZ/2R \quad \beta' \equiv kbY/2R$$



Irradiance:

$$I(Y, Z) = I(0) \left(\frac{\sin \alpha'}{\alpha'} \right)^2 \left(\frac{\sin \beta'}{\beta'} \right)^2$$

$I(0)$: the irradiance at $Y=0$ and $Z=0$.



The circular aperture

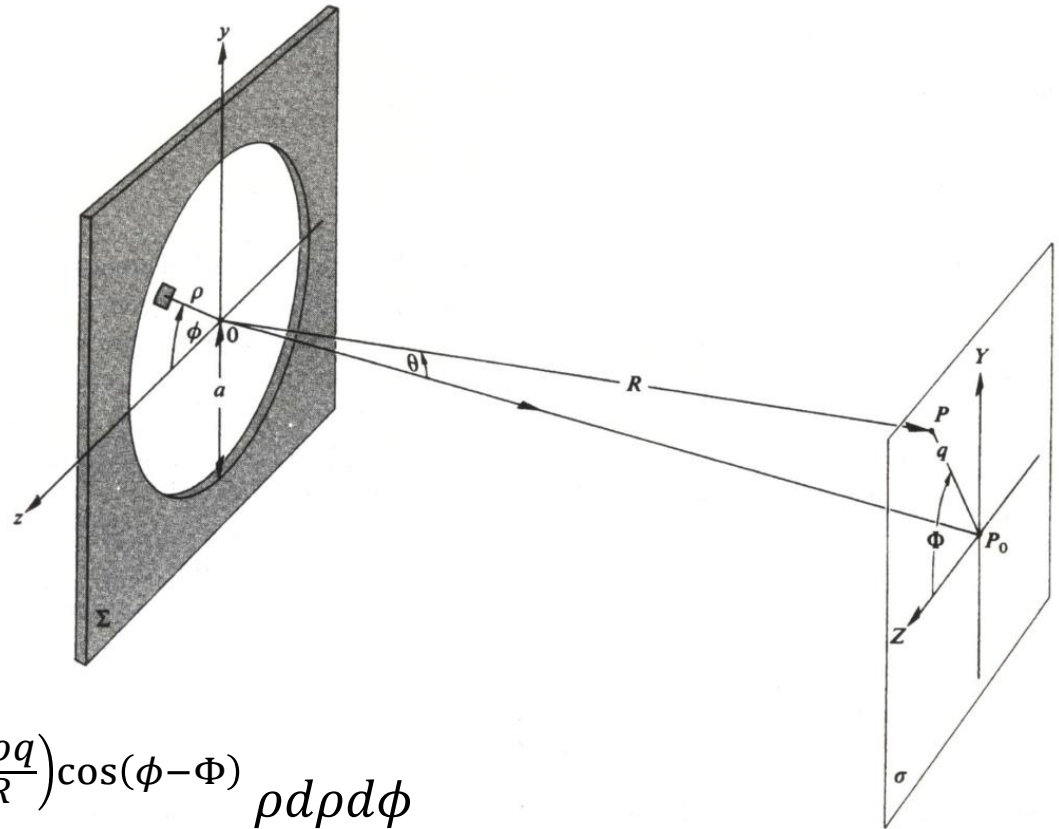
Optical disturbance at P:

$$\tilde{E} = \frac{\epsilon_A e^{i(\omega t - kR)}}{R} \int_{\text{Aperture}} e^{ik(Yy + Zz)} dS$$

For a circular opening:

$$z = \rho \cos \phi \quad y = \rho \sin \phi$$

$$Z = q \cos \Phi \quad Y = q \sin \Phi$$



$$\tilde{E} = \frac{\epsilon_A e^{i(\omega t - kR)}}{R} \int_{\rho=0}^a \int_{\phi=0}^{2\pi} e^{i\left(\frac{k\rho q}{R}\right)\cos(\phi - \Phi)} \rho d\rho d\phi$$

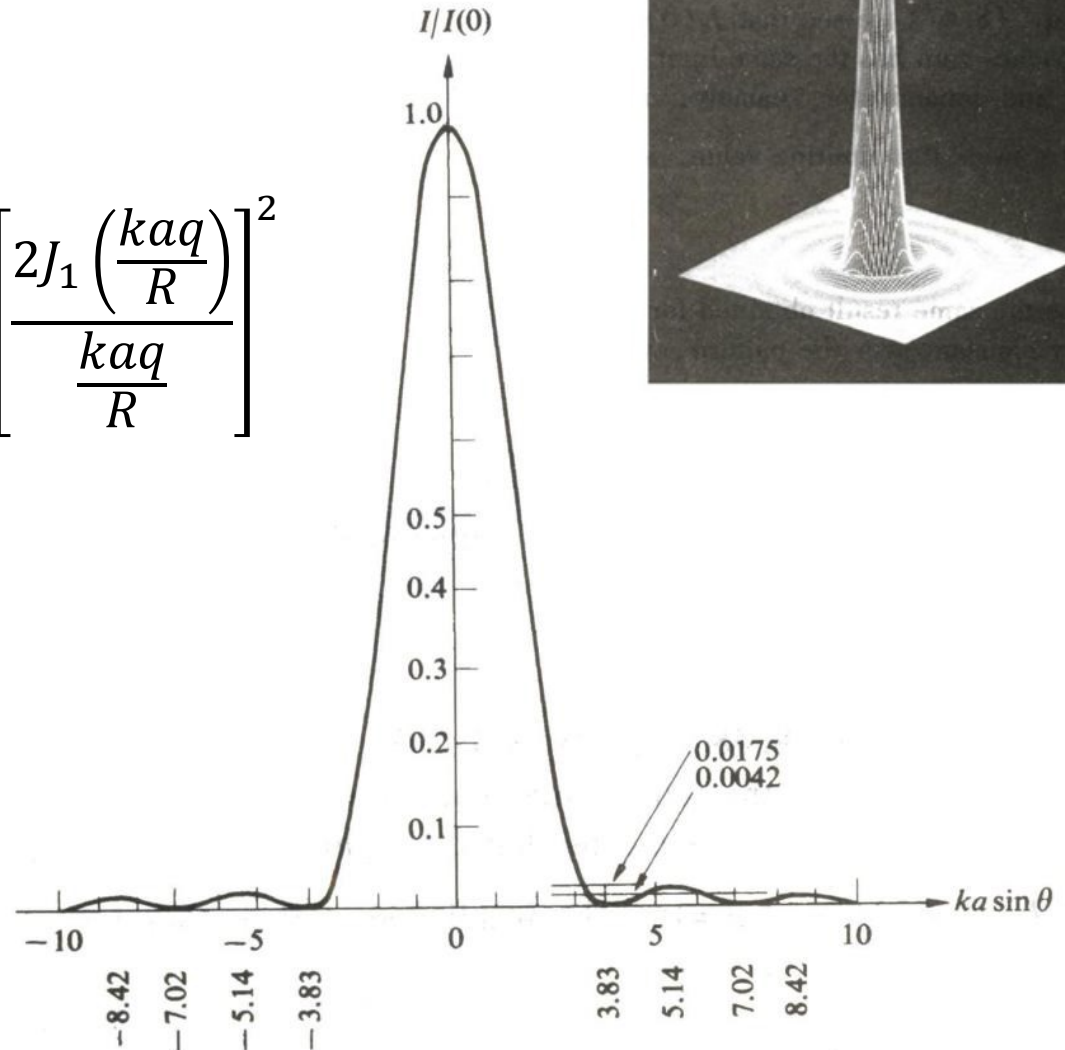
$$\tilde{E} = \frac{\epsilon_A e^{i(\omega t - kR)}}{R} 2\pi a^2 \left(\frac{R}{kaq} \right) J_1 \left(\frac{kaq}{R} \right)$$

The irradiance at P:

$$I = \frac{2\epsilon_A^2 A^2}{R^2} \left[\frac{J_1 \left(\frac{kaq}{R} \right)}{\frac{kaq}{R}} \right]^2 = I(0) \left[\frac{2J_1 \left(\frac{kaq}{R} \right)}{\frac{kaq}{R}} \right]^2$$

$$= I(0) \left[\frac{2J_1(ka \sin \theta)}{ka \sin \theta} \right]^2$$

$I(0)$: the irradiance at P_0 .



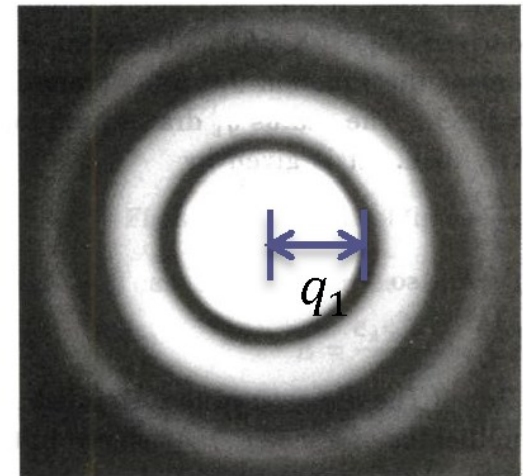
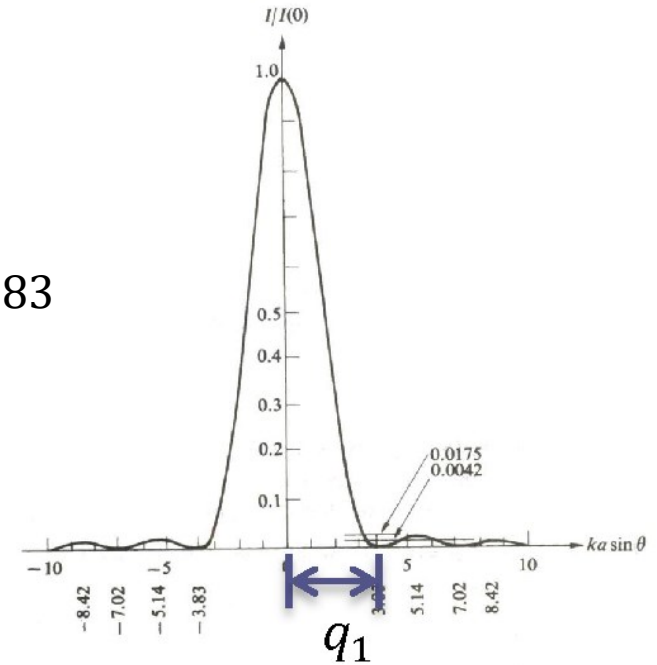
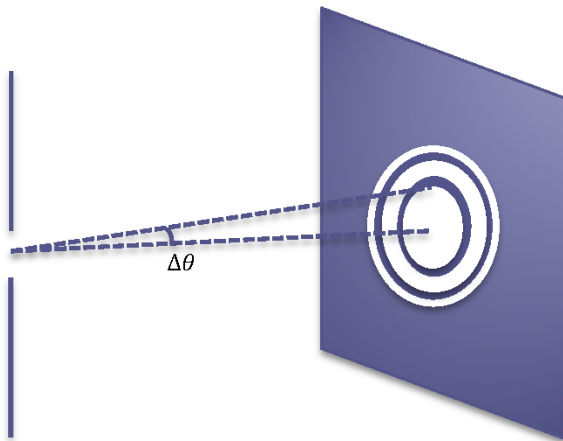
Airy disk: the towering central maximum corresponds to a high-irradiance circular spot.

The radius of Airy disk: $J_1(u) = 0$ when $u = \frac{kaq}{R} = 3.83$

$$q_1 = \frac{3.83 R}{ka} = 1.22 \frac{R\lambda}{2a} \approx 1.22 \frac{f\lambda}{D}$$

The angular measure:

$$\Delta\theta \approx \sin \Delta\theta = \frac{q_1}{f} = 1.22\lambda/D$$



Airy rings using a 0.5-mm hole diameter

Resolution of Imaging Systems

Example: two stars seen through the objective lens of a telescope (the entrance pupil corresponds to the diffracting aperture)

If Lord Rayleigh's criterion is applied, the stars are said to be just resolved when the center of one Airy disk falls on the first minimum of the Airy pattern of the other star.

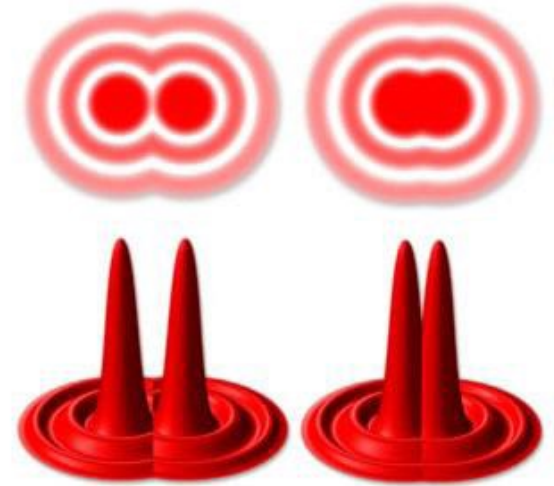
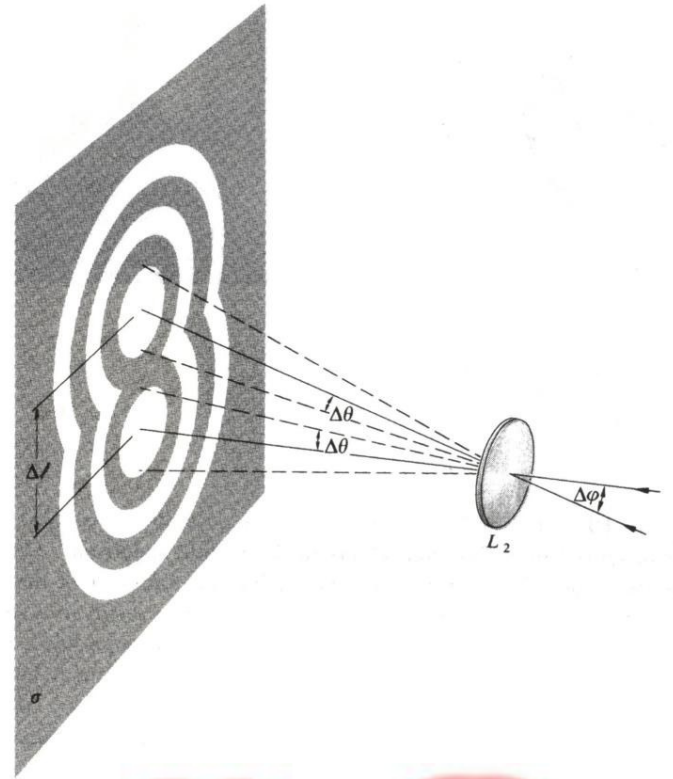
The minimum resolvable angular separation (Lord Rayleigh's criterion): 瑞利判据

$$(\Delta\varphi)_{min} = \Delta\theta = 1.22\lambda/D$$

← 在非空气介质中:
 $n\Delta\theta = 1.22\lambda/D$

Improve the resolution:

- ☐ Decrease the wavelength;
- ☐ Increase the diameter of the objective lens.



Lord Rayleigh's criterion

Grating Spectroscopy 得到周期性结果: 干涉

The angular distance between the zeros on either side of a principal maximum.

$$\Delta\alpha = 2\pi/N$$

At oblique incidence:

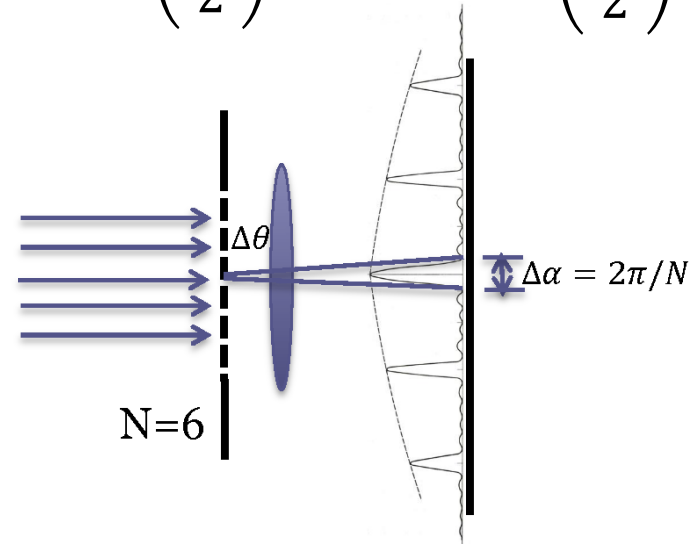
$$\alpha = (ka/2)(\sin\theta - \sin\theta_i)$$

$$\Delta\alpha = \left(\frac{ka}{2}\right) \cos\theta (\Delta\theta) = 2\pi/N$$

$$I(\theta) = I_0 \left(\frac{\sin\beta}{\beta}\right)^2 \left(\frac{\sin N\alpha}{\sin\alpha}\right)^2$$

用光敏性材料记录光强信息 → holography 全息术

$$\beta \equiv \left(\frac{kb}{2}\right) \sin\theta \quad \alpha \equiv \left(\frac{ka}{2}\right) \sin\theta$$



Angular width of a line: $\Delta\theta = 2\lambda/(Na \cos\theta_m)$

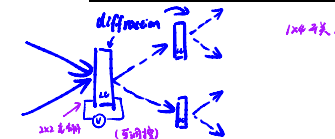
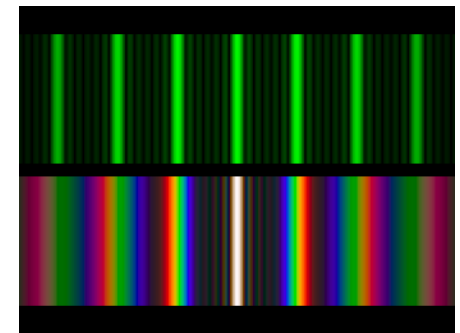
↙ Grating width

Angular dispersion: the difference in angular position corresponding to a difference in wavelength:

$$\frac{d\theta}{d\lambda} = \frac{m}{a \cos\theta_m}$$

$$a(\sin\theta_m - \sin\theta_i) = m\lambda$$

$$a \cos\theta d\theta = m d\lambda$$



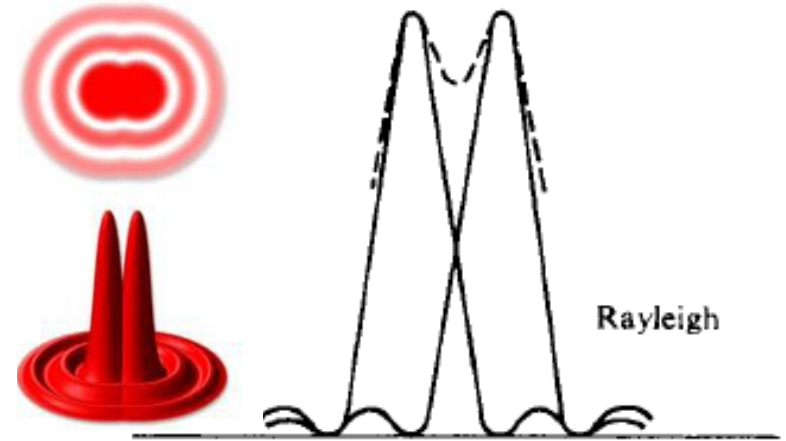
瑞利判据

Lord Rayleigh's criterion: the principal maximum of one coincide with the first minimum of the other, at the limit of resolution, the angular separation is half the linewidth, $d\theta = \frac{\Delta\theta}{2}$

$$(\Delta\theta)_{\min} = \lambda / (Na \cos \theta_m)$$

$$\frac{d\theta}{d\lambda} = \frac{m}{a \cos \theta_m}$$


$$(\Delta\theta)_{\min} = (\Delta\lambda)_{\min} m / (a \cos \theta_m)$$



Lord Rayleigh's criterion

The chromatic resolving power (色分辨本领) of a spectrometer:

$$\mathcal{R} \equiv \frac{\lambda}{(\Delta\lambda)_{\min}} = mN = \frac{Na(\sin\theta_m - \sin\theta_i)}{\lambda}$$

1/mm $\rightarrow$ 1,000,000 lines

Example: A grating 6 inches wide and containing 15,000 lines per inch will have a total of 9×10^4 lines and a resolving power, in the second order, of $R = 1.8 \times 10^5$. In the vicinity of 540 nm, the grating could resolve a wavelength difference of 0.003 nm.

9.3 Fresnel Diffraction

(菲涅尔对惠更斯原理的改良)

Huygens-Fresnel Principle: at any instant, every point on the primary wavefront is envisioned as a continuous emitter of spherical secondary wavelet. It means each wavelet radiate uniformly in all direction.

In experiment, there is no reverse wave traveling back toward the source.

Obliquity (inclination factor): it describes the directionality of the secondary emissions.

Kirchhoff formulation: 基尔霍夫公式

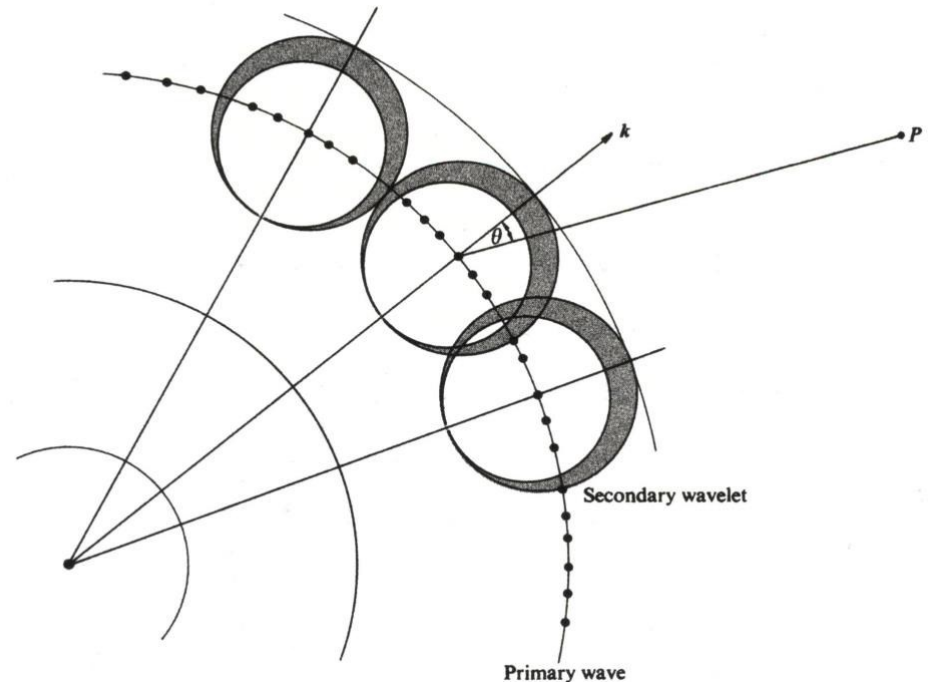
$$K(\theta) = \frac{1}{2}(1 + \cos \theta)$$

倾斜因子

θ : the angle made with the normal to the primary wavefront (k).

人为消除反向波.

向前 $\theta=0$: $K=1$ 向后 $\theta=180^\circ$: $K=0$.



Case: the free propagation of a spherical monochromatic wave emitted from a point source, S.

The spherical surface: primary wavefront at some arbitrary time t' emitted from S at $t=0$.

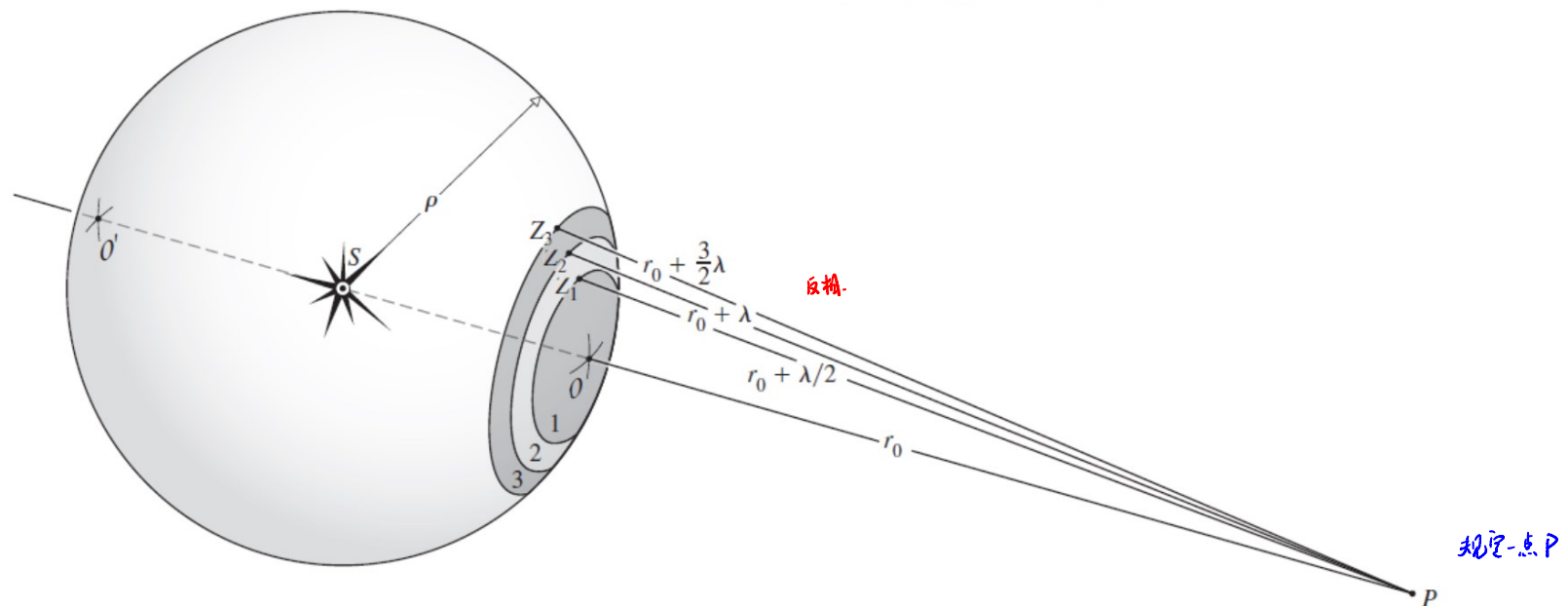
The disturbance:

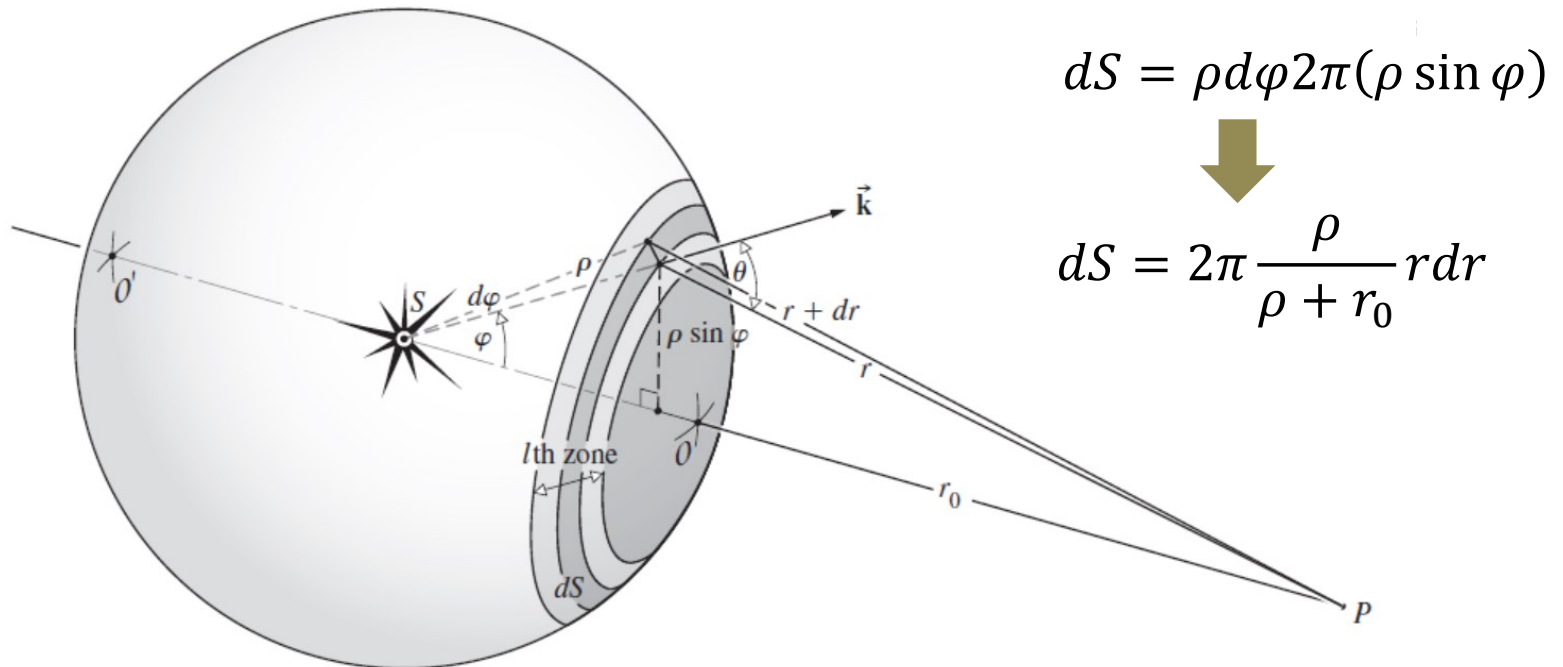
菲涅尔波带片

$$E = \frac{\epsilon_0}{\rho} \cos(\omega t' - k\rho)$$

波源强度.

Fresnel (half-period zones): divide the wavefront into a number of annular region with the radius of $r_0 + \lambda/2, r_0 + \lambda, r_0 + 3\lambda/2, \dots$, centered at P.





The disturbance arriving at P from the l^{th} zone:

$$E_l = K_l 2\pi \frac{\epsilon_A \rho}{\rho + r_0} \int_{r_{l-1}}^{r_l} \cos[\omega t - k(\rho + r)] dr$$

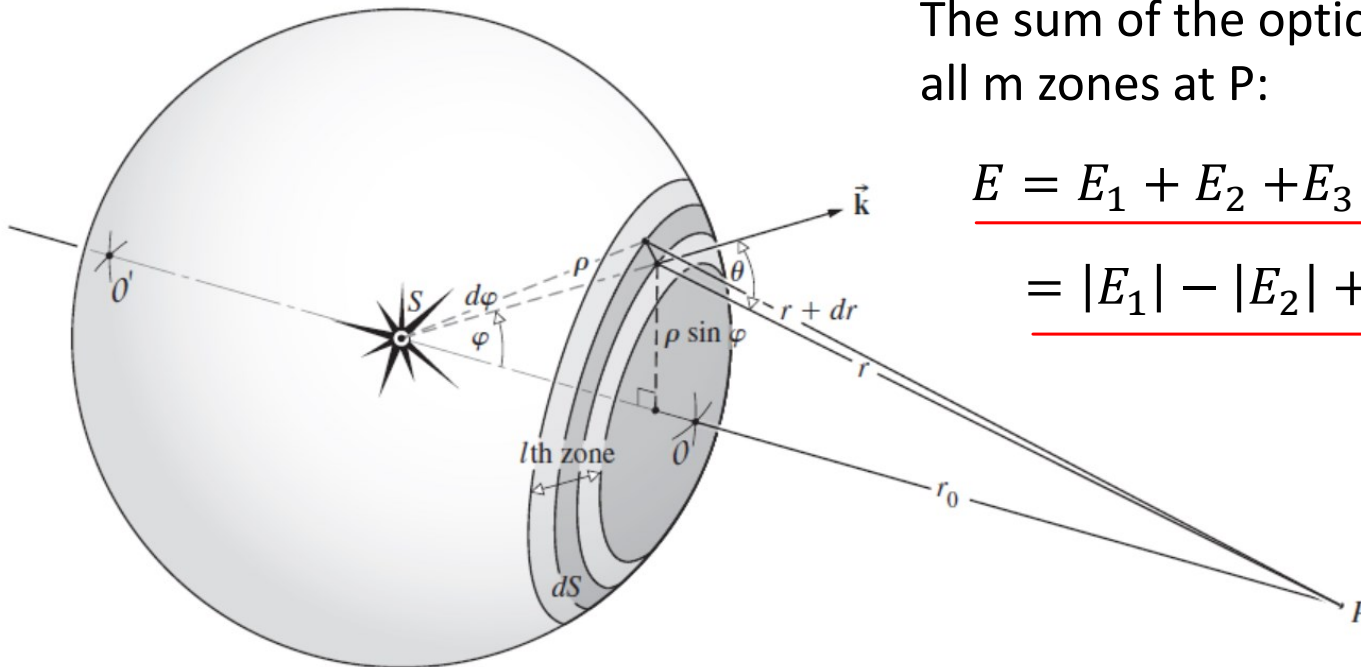
$$r_l = r_0 + l\lambda/2$$

$$r_{l-1} = r_0 + (l-1)\lambda/2$$

$$E_l = (-1)^{l+1} \frac{2K_l \epsilon_A \rho \lambda}{(\rho + r_0)} \sin[\omega t - k(\rho + r_0)]$$

The sum of the optical disturbances from all m zones at P:

$$\begin{aligned} E &= E_1 + E_2 + E_3 + \cdots + E_m \\ &= |E_1| - |E_2| + |E_3| - \cdots + |E_m| \end{aligned}$$



Case I. if m is odd, the sum of the optical disturbances from all m zones at P:

$$E = |E_1| - |E_2| + |E_3| - \cdots + |E_m|$$

$$E = \frac{|E_1|}{2} + \left(\frac{|E_1|}{2} - |E_2| + \frac{|E_3|}{2} \right) + \left(\frac{|E_3|}{2} - |E_4| + \frac{|E_5|}{2} \right) + \cdots + \left(\frac{|E_{m-2}|}{2} - |E_{m-1}| + \frac{|E_m|}{2} \right) + \frac{|E_m|}{2} \quad (I)$$

$$E = |E_1| - \frac{|E_2|}{2} - \left(\frac{|E_2|}{2} - |E_3| + \frac{|E_4|}{2} \right) - \left(\frac{|E_4|}{2} - |E_5| + \frac{|E_6|}{2} \right) + \cdots + \left(\frac{|E_{m-3}|}{2} - |E_{m-2}| + \frac{|E_{m-1}|}{2} \right) - \frac{|E_{m-1}|}{2} + |E_m| \quad (II)$$

If $|E_l| > (|E_{l-1}| + |E_{l+1}|)/2$

Formula I: $E < \frac{|E_1|}{2} + \frac{|E_m|}{2}$

Formula II: $E > |E_1| - \frac{|E_2|}{2} - \frac{|E_{m-1}|}{2} + |E_m| \xrightarrow{|E_1| \approx |E_2|, |E_{m-1}| \approx |E_m|} \frac{|E_1|}{2} + \frac{|E_m|}{2}$

→ (本质上相同)

$$E \approx \frac{|E_1|}{2} + \frac{|E_m|}{2}$$

If $|E_l| < (|E_{l-1}| + |E_{l+1}|)/2$

Formula I: $E > \frac{|E_1|}{2} + \frac{|E_m|}{2}$

Formula II: $E < |E_1| - \frac{|E_2|}{2} - \frac{|E_{m-1}|}{2} + |E_m| \xrightarrow{|E_1| \approx |E_2|, |E_{m-1}| \approx |E_m|} \frac{|E_1|}{2} + \frac{|E_m|}{2}$

$$E \approx \frac{|E_1|}{2} + \frac{|E_m|}{2}$$

Case II: if m is even, the sum of the optical disturbances from all m zones at P :

$$E = |E_1| - |E_2| + |E_3| - \dots - |E_m| \approx \frac{|E_1|}{2} - \frac{|E_m|}{2} \quad \text{homework}$$

The sum of the optical disturbances

$$E \approx \begin{cases} \frac{|E_1|}{2} + \frac{|E_m|}{2} & m \text{ is odd} \\ \frac{|E_1|}{2} - \frac{|E_m|}{2} & m \text{ is even} \end{cases}$$

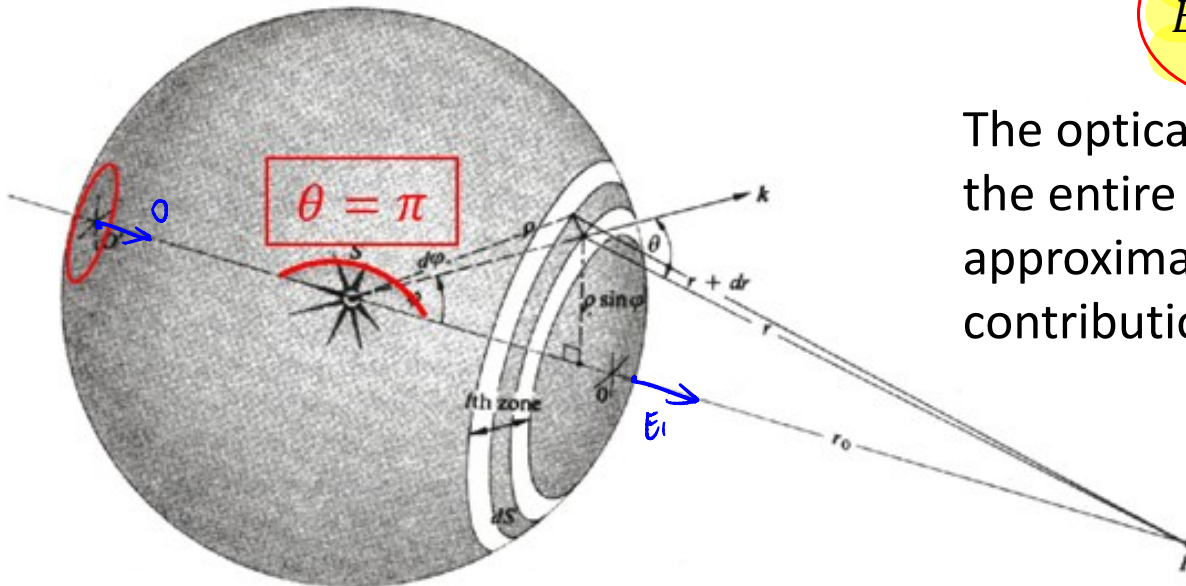
Kirchhoff's correct obliquity:

$$K_m(\pi) = 0, |E_m| = 0$$

So, we have

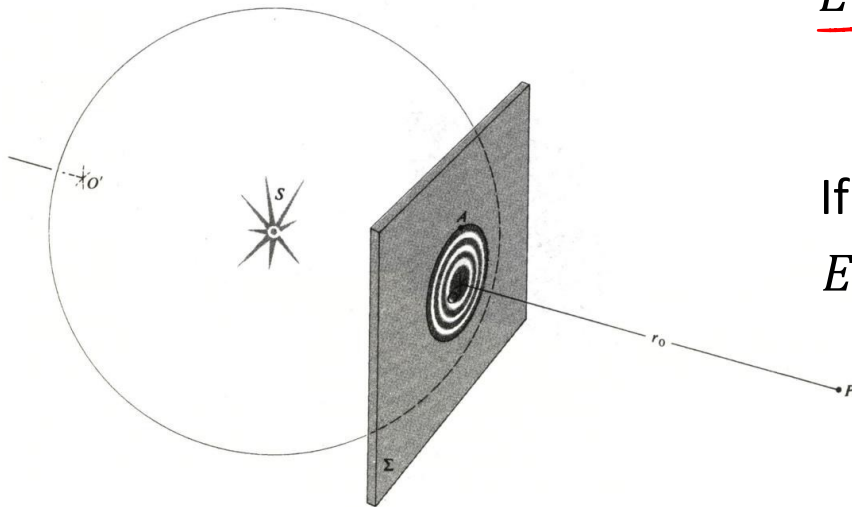
$$E \approx \frac{|E_1|}{2}$$

The optical disturbance generated by the entire unobstructed wavefront is approximately equal to one-half of the contribution from the first zone.



Circular Apertures

A monochromatic spherical wave impinging on a screen containing a small hole.



The total disturbance at P , if m is even:

$$\underline{E} = (|E_1| - |E_2|) + (|E_3| - |E_4|) + \dots \\ + (|E_{m-1}| - |E_m|) \approx \underline{0}$$

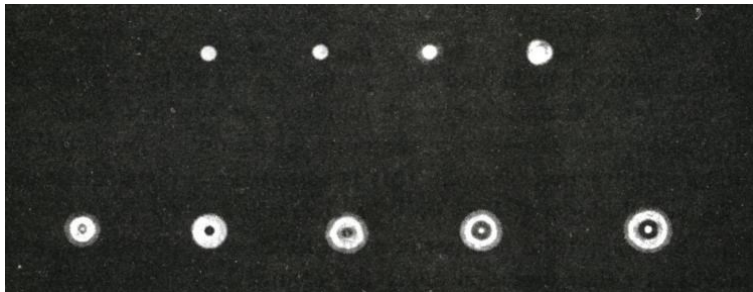
If m is odd

$$E = |E_1| - (|E_2| - |E_3|) - (|E_4| - |E_5|) - \dots \\ - (|E_{m-1}| - |E_m|) \approx |E_1|$$

Note:

it is roughly twice the amplitude of the unobstructed wave

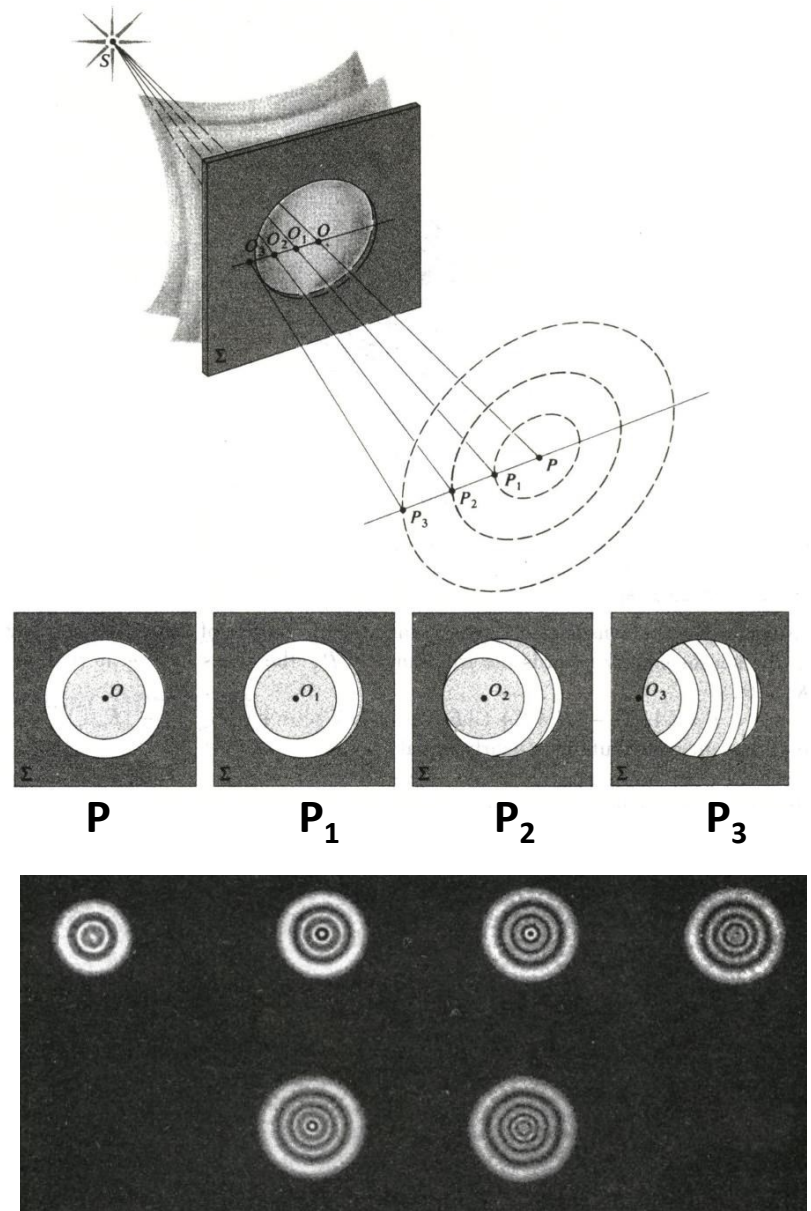
If m is not a integer, the irradiance at P is somewhere between zero and its maximum value



Diffraction patterns for circular apertures of increasing size

The rest of the pattern: assume that two complete zones fill the aperture

- At P: $E \approx 0$
- At P₁: the second zone has been partially obscured and the third begins to show. $E \neq 0$
- At P₂: a good fraction of the second zone is hidden, the third is even more evident. The contribution from the first and third zones are in-phase. A bright spot
- At P₃: the first zone is partially uncovered. The last few contributing segments is negligible. Further into the shadow, the entire first zone is obscured, the last terms are negligible. Darkness



Diffraction patterns for circular apertures of increasing size

The number of zones in a given aperture

The area of each zone:

$$dS = 2\pi \frac{\rho}{\rho + r_0} r dr \Rightarrow A = \frac{\rho}{(\rho + r_0)} \pi r_0 \lambda$$

The number zones within the aperture with a radius R.

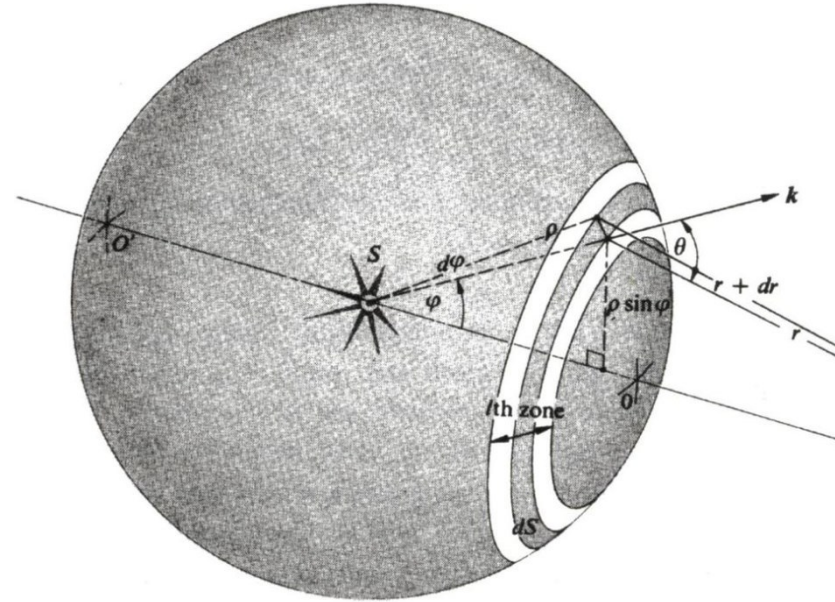
$$N = \frac{\pi R^2}{A} = \frac{(\rho + r_0) R^2}{\rho r_0 \lambda}$$

Example: a point source 1m behind the aperture ($\rho \approx 1m$), a plane of observation 1m in front of aperture ($r_0 = 1m$).

$$\lambda = 500 \text{ nm}$$

$$R = 1 \text{ mm}; 4 \text{ zones}$$

$$R = 1 \text{ cm}; 400 \text{ zones}$$



$$N \propto \frac{1}{r_0}$$

As P moves in either direction along the central axis, the number of uncovered zones oscillates between odd and even integers. As a result, the irradiance goes through a series of maxima and minima.

Circular Obstacles

When a circular obstacle is inserted between a light source and a screen P, suppose the first l zones are blocked off. The electrical field at P:

$$E = |E_{l+1}| - |E_{l+2}| + \cdots \pm |E_m|$$
$$\approx \frac{|E_{l+1}|}{2} \pm \frac{|E_m|}{2}$$

$K_m \rightarrow 0$, E_m approaches zero.

$$E \approx \frac{|E_{l+1}|}{2}$$

The irradiance on the central axis is only slightly less than that of the unobstructed wave ($E \approx \frac{|E_1|}{2}$).

There is a bright spot everywhere along the central axis except immediately behind the circular obstacle (because θ increases, $K_m \rightarrow 0$).

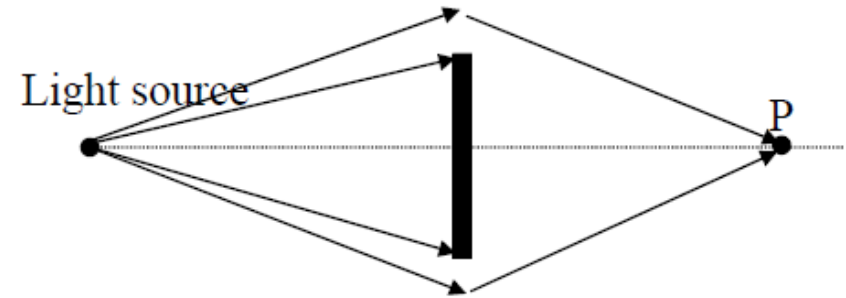
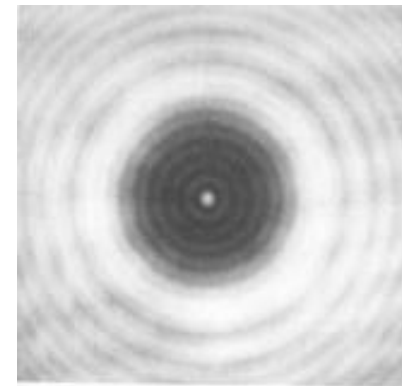
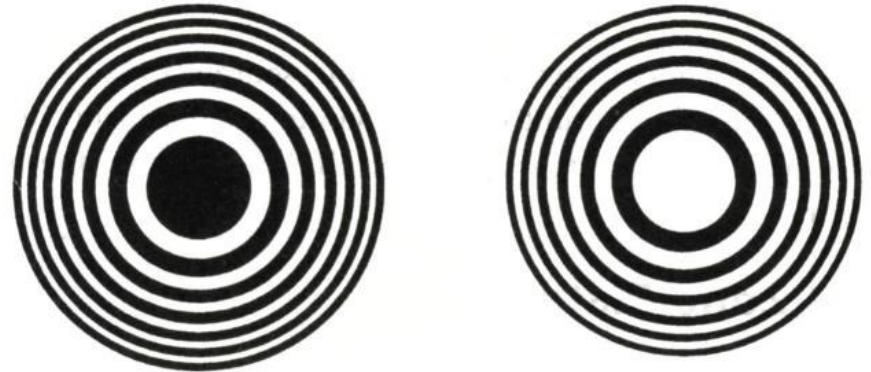


Fig 8.20 Circular Obstacle



The Fresnel Zone Plate

Zone plate: a screen that alter light, either in amplitude or phase, coming from every other half-period zone.



Example: a zone plate that passes only the first 20 odd zones and obstructs the even zone.

$$E = E_1 + E_3 + E_5 + \cdots + E_{39}$$

The disturbance at P: $E \approx 20E_1$

For an unobstructed wavefront: $E \approx \frac{1}{2}E_1$

The irradiance has been increased by a factor of 1600.

Calculate the radii of the zones

A wave traveling the path $S-Am-P$ must have the optical path difference by $m\lambda/2$ with a wave that travels the path $S-O-P$. (Am is the outer edge of the m th zone)

$$(\rho_m + r_m) - (\rho_0 + r_0) = m\lambda/2$$

Because

$$\rho_m = (R_m^2 + \rho_0^2)^{1/2} \quad r_m = (R_m^2 + r_0^2)^{1/2}$$

Expand in binomial series

$$\rho_m = \rho_0 + \frac{R_m^2}{2\rho_0} \quad r_m = r_0 + \frac{R_m^2}{2r_0}$$

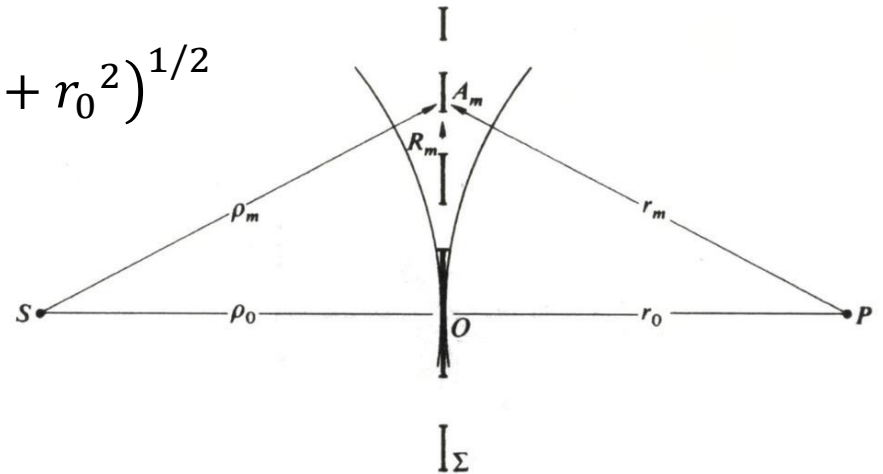
We have $\left(\frac{1}{\rho_0} + \frac{1}{r_0} \right) = \frac{m\lambda}{R_m^2}$

菲涅尔透镜

When $\rho_0 \rightarrow \infty$, $R_m^2 = mr_0\lambda$ which has the same form of thin lens equation.

$$f_1 = \frac{R_m^2}{m\lambda}$$

The S and P are conjugate foci, but there are several focal points



Homework

□ 10.32, 10.35, 10.37